

Government Reputation, FDI, and Profit Shifting*

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Abstract

This paper documents that a higher government reputation is associated with higher statutory corporate tax rates, greater inward foreign direct investment, and more net inward profit shifting, conditional on country characteristics. I develop a model in which firms invest before the government implements its tax but can shift profits afterward. Government reputation is modeled as the belief of multinational firms that the government can commit to its tax policy. Without profit shifting, a better reputation expands investment sufficiently that the government chooses a higher tax under commitment. I analytically show that profit shifting can reverse this result. By protecting returns against discretionary taxation, profit shifting sustains investment even under a poor reputation and reduces the expansion of the investment base that a better reputation alone could provide. When this protection is sufficiently strong, the greater responsiveness of investment to a credible tax reduction induces the government to choose a lower rate. Investment and revenue rise despite the lower tax. In a dynamic extension, reneging reduces investment by future firms. Higher reputation increases the future revenue loss from reneging, which induces the government to sustain a lower tax rate.

Keywords: Commitment, Corporate taxation, Foreign direct investment, Government reputation, Multinational firm, Profit shifting

JEL codes: H21, H25, H26, F21, F23

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1 Introduction

A government's reputation for maintaining its tax policy matters for foreign investment because a time-inconsistent government can raise taxes on profits after firms have invested. When firms anticipate such a revision, they invest less than they would under a government that can commit to its tax policy. This paper studies optimal corporate taxation in an open economy where multinational firms do not know whether the government has commitment. Firms have another way to respond to a tax increase besides changing investment. They can shift profits across countries once the tax is realized. I examine how this option changes the relationship between reputation and the optimal corporate tax rate.

How does reputation affect the government's choice of tax? A better reputation may allow the government to impose a high tax rate by attracting firms' investment and expanding the tax base. At the same time, it may lower the optimal rate because it amplifies the distortion of a marginal tax increase on investment. Consider a reduction in the initial tax. Firms invest little in response if they believe the government will raise it again after their capital is installed. A government with a better reputation can obtain a larger investment response to the same reduction. Whether it chooses to cut the tax depends on the relative importance of these two effects. I find that the optimal rate increases with reputation without profit shifting. The rate can instead decrease when shifting allows firms to protect enough of their profits from an opportunistic government.

I begin by documenting three empirical facts about the quality of government, corporate taxes, and multinational firms. The quality of government captures administrative continuity, legal enforcement, and control of corruption, which can help firms assess the credibility of government policy. First, higher quality of government is associated with higher statutory corporate tax rates across countries. Using country means over 2003–2021, I estimate the association after controlling for income, population, and resource dependence. The result is reflected in both Tax Foundation data and the combined statutory rates of the Organization for Economic Co-operation and Development (OECD). The estimates also remain positive when the two sources are restricted to the same country-year observations. Second, I find a positive within-country association between the quality of government and inward foreign direct investment (FDI). I use the annual direct-investment inflows of [Forbes and Warnock \(2021\)](#) in logs and control for country and year effects. With the additional economic controls, a 0.1 increase in the prior year's quality-of-government index is associated with 0.213 log points more inflows.

Third, a higher quality of government is associated with greater net inward profit shifting, conditional on the statutory tax rate. Using the 2016 estimates of [Garcia-Bernardo and Jansky \(2021\)](#), I find that a 0.1 higher in the quality-of-government index corresponds to approximately one percentage point of gross domestic product (GDP) more net inward shifting. Countries with negative net inward shifting are retained in the sample. The reporting decision can be made later than the investment decision. [Hopland et al. \(2018\)](#) show that multinational firms adjust intragroup payments within the tax year, while [de Mooij and Liu \(2020\)](#) find lower investment by multinational affiliates relative to domestic firms following transfer-pricing regulation. I use these observations to motivate firms' choices regarding profit transfers after investment in the model.

I also study investment around the 2018 *Achmea* judgment, which restricted investor-state arbitration under treaties between European Union (EU) countries. The analysis follows tangible investment by foreign affiliates in 22 host and parent countries during 2010–2019. I compare pairs exposed to the judgment through their treaty coverage with pairs without such coverage, allowing for pair, host-year, and parent-year fixed effects. Investment by the exposed pairs falls after the judgment. This result continues to hold with pair-specific linear trends. In that specification the pre-event coefficients are jointly insignificant. Thus the evidence includes both the associations with quality of government and a change in the enforcement of investment protection.

I set up a small open economy with a government and multinational firms that maximize the sum of profits across the host country and the rest of the world. Government type is initially unknown to firms. The commitment type must implement its initial tax, while the opportunistic type reoptimizes after firms make irreversible investments. Firms therefore invest according to the expected tax treatment under the two types, weighted by prior reputation. They observe the tax before deciding on profit transfers. I solve for the commitment type's preferred initial rate among those that both types are willing to announce.

The one-period model gives opposite tax responses depending on whether firms can shift profits. In the absence of shifting, an opportunistic government takes all host profits. Firms' willingness to invest then depends heavily on their belief that the government has commitment. A better reputation raises investment enough for the commitment type to choose a higher tax rate. This is the case that motivates the positive association between quality of government and statutory taxes in the data. With shifting, however, firms retain profits even if the government reoptimizes. Their investment is consequently less dependent on reputation, while remaining responsive to the initial tax rate. The commitment type can gain more revenue by lowering that

rate as reputation improves. I prove the decrease for all positive values of reputation when firms can protect a sufficiently large share of their return under discretion.

To obtain analytical results, I let firms deduct shifting costs in the country from which they transfer profits. The numerical analysis allows deductions in both countries. The optimal commitment tax again decreases in reputation, from 45.2 percent at a commitment probability of 0.1 to 35.9 percent at 0.9 in the baseline. Both investment and revenue increase over this range. The tax decrease is also present with alternative cost functions, foreign tax rates, and heterogeneous shifting costs. I separately examine limits on the amount firms can transfer. The relevant limit cannot be inferred from the amount they actually shift. In one numerical example, only 24.2 percent of host profits is shifted under discretion. Yet firms must be able to transfer more than 55.6 percent to deter the government from taxing at the confiscatory rate. I provide an analytical expression for this threshold.

In the two-period model, reneging on the first-period tax also changes what future cohorts of firms believe about the government. Both types maintain the initial tax in the equilibrium I consider. Were the opportunistic type to revise it, the next cohort would know its type and invest less. This loss of future revenue is greater for a government with a better initial reputation. The government is thus willing to implement a lower tax as reputation improves, until the full-commitment rate is attainable. First-period investment increases as a result. The comparison extends to any finite number of periods with successive cohorts of firms. In numerical exercises, I also find that higher shifting costs increase the immediate gain from reneging more than the discounted future loss. The sustainable rate rises by more than the full-commitment rate, particularly at a low initial reputation.

The paper contributes to the literature on tax-system design by relating restrictions on avoidance to government commitment. The collection technology in [Slemrod \(1990\)](#) and the administrative instruments in [Slemrod and Kopczuk \(2002\)](#) affect which behavioral responses matter for the optimal tax. My transfer-limit result concerns a government choosing its tax after investment has been sunk. At the initial rate, neither the base nor its derivative need change when the limit is imposed. Nevertheless, confiscation can become optimal. A local elasticity at that rate therefore leaves out a reporting response relevant to the choice of revision.

This paper also relates to the literature on international tax avoidance and competition ([Peralta et al., 2006](#); [Devereux et al., 2008](#); [Johannesen, 2010](#)). Tax planning supports investment through a lower effective burden in [Hong and Smart \(2010\)](#). [Slemrod and Wilson \(2009\)](#) study its resource costs and fiscal externalities, while [Mongrain et al. \(2023\)](#) explain how restrictions affect investment and retained-profit responses in opposite directions. I introduce the government's decision to maintain or revise the initial

rate. I can then compare a change in shifting costs when implementation is guaranteed with the same change when it must satisfy an incentive constraint. In the numerical model, limited commitment adds to the increase in the tax, with a larger additional increase at lower reputation.

The theory builds on work on time-consistent capital taxation ([Chari et al., 1988](#)). [Lu \(2013\)](#) studies privately known commitment in tax announcements, and [Dovis and Kirpalani \(2021\)](#) analyze policy rules when commitment is uncertain. Studies of expropriation establish how the loss of future investment restrains a government that taxes installed capital ([Thomas and Worrall, 1994](#); [Aguiar and Amador, 2011](#); [Schnitzer, 1999](#)). In my setting, firms can protect some of their profits even after investment. This changes the optimal tax under commitment and also affects investment after reputation has been lost. I account for both when deriving a first-period rate that both types implement.

Firms' responses to the taxation of sunk investment have been studied through several mechanisms. Excess production capacity abroad disciplines taxation in [Janeba \(2000\)](#). Private information about intrafirm trading costs protects information rents in [Konrad and Lommerud \(2001\)](#). Firm mobility enters tax choices in [Langenmayr and Simmler \(2021\)](#), while [Langenmayr and Simmler \(2024\)](#) find that the anticipation of higher future taxes discourages entry. I consider firms whose capital is fixed within a period but whose reported profits remain adjustable. This reporting choice determines the return they can preserve under discretion and, in turn, how reputation affects the optimal tax.¹

[Mardan and Stimmelmayer \(2020\)](#) is the closest comparison on risk and corporate taxes. They find that shifting costs can reverse the tax response to country risk, where risk is a probability of project failure. Risk in my model comes from the government's own tax decision after investment. Only the government knows whether it has commitment, and a revision reveals this information to firms investing in later periods. The tax that maximizes revenue under full commitment may then fail to be implemented by an opportunistic type. I solve for the tax it will implement and compare it with the full-commitment choice. This incentive to revise policy is absent from a tax choice under exogenous project risk.

The paper is organized as follows. Section 2 presents the empirical facts. Section 3 introduces the model, and Section 4 analyzes optimal taxation under commitment and discretion. Section 5 studies sustainable taxation when policy affects future investment. Section 6 concludes.

¹Investment can also be protected through international agreements ([Kohler and Stähler, 2019](#); [Horn and Tangerås, 2021](#)). The government's cost of revising policy in my dynamic model comes from the response of subsequent cohorts of firms.

2 Empirical Facts

I use the International Country Risk Guide (ICRG) Quality of Government (QoG) index to examine corporate tax rates, inward FDI, and the location of reported profits. The comparisons distinguish the tax policy chosen by the government from firms' investment and reporting decisions.

2.1 Quality of Government and Tax Credibility

A firm's return on foreign investment depends partly on how the host government implements its policies once capital is installed. Administrative continuity makes that treatment more predictable, while legal constraints can limit a government's discretion. This is the connection between institutions and credible policy emphasized by [North and Weingast \(1989\)](#). It also motivates the distinction made by [Acemoglu and Johnson \(2005\)](#) between constraints on government and the enforcement of private contracts. My concern is with constraints on the government's treatment of firms after they have invested.

I use ICRG Quality of Government to measure the institutional support for credible policy implementation. The index, as defined by the Quality of Government Institute ([Dahlberg et al., 2026](#)), averages three scores, Corruption, Law and Order, and Bureaucracy Quality. I reconstruct it from the original monthly observations, dividing the first two scores by six and the third by four. Equal weights give an index between zero and one, increasing in the quality of government. I average all twelve months to obtain each annual observation and use the same index in all three empirical exercises.

The components have an interpretation relevant for policy implementation ([The PRS Group, nd](#)). Bureaucracy Quality scores administrative expertise and independence from political pressure, including whether administration continues when the government changes. Law and Order combines the strength and impartiality of the legal system with compliance with the law. Corruption covers patronage and private influence on public decisions, with payment demands in tax assessment among the practices considered. These features of government help firms judge whether they can rely on the government's established policies when making investment and reporting decisions.

Government reputation in the model is firms' belief that the government can commit to its tax policy. A credible tax policy gives firms a reason to expect that the rate chosen before investment will govern the taxation of their subsequent profits. Administrative continuity and constraints on discretionary action can support that expectation, which

is why I use QoG as a proxy for the institutional basis of reputation. The index does not give a direct probability of commitment.²

In both the tax and FDI exercises, QoG is measured in the year before the outcome. I compare countries' average tax rates, estimate the FDI relation using annual changes within countries, and use a single cross section for profit shifting. Sources and construction details are in Appendix C.1.

2.2 Quality of Government, Taxation, and FDI

I assemble statutory rates including both central and subnational taxation from Tax Foundation (Enache, 2022) and OECD (OECD, 2026). For each source, I retain country-year observations with a tax rate, the previous year's QoG, and all controls. I then average each variable within country over these observations for 2003–2021. This avoids averaging the tax and the regressors over different years. I estimate the following country-level regression

$$\overline{\text{Tax rate}}_h = \alpha + \beta \overline{\text{QoG}}_h + \delta' \overline{\text{Controls}}_h + u_h. \quad (1)$$

Controls are log real GDP per capita, log population, and the share of natural-resource rents in GDP, accounting for development, market size, and resource dependence. Each country receives one observation with equal weight. Reported standard errors are heteroskedasticity robust with an HC3 correction, which adjusts for the leverage of each observation. All countries with the tax and these regressors can enter. I do not condition inclusion on coverage in the Forbes–Warnock investment data.

Both sources give a positive coefficient on QoG in Table 1. A 0.1 increase in the index is associated with a tax rate higher by 0.884 percentage points using Tax Foundation data and 1.777 points using OECD data. The respective p -values are 0.058 and 0.006. Matching the country-years across sources gives increases of 1.333 and 1.646 points. I also exclude tax havens and add continent effects in separate regressions. In each case, the coefficient remains positive and significant at the 10 percent level. All of these comparisons control for income, population, and resource dependence. I report the additional estimates and discuss the variation used to obtain them in Appendix C.2.

My measure of FDI is the inward direct-investment flow in Forbes and Warnock (2021), summed over the four quarters of each year. I use logs and therefore retain positive annual totals. In this series, foreign direct investors' withdrawals are deducted from their equity investments, reinvested earnings, and intercompany lending in the

²Quality of government may also affect investment and taxation through public services and tax administration. The model holds production and reporting technologies fixed to examine the role of reputation.

TABLE 1: Quality of government and statutory corporate tax rates

	(1)	(2)	(3)	(4)
Quality of government	8.844* (4.615)	17.767*** (6.328)	13.326** (6.192)	16.459*** (6.145)
Source	TF	OECD	TF	OECD
Country-year coverage	Available	Available	Matched	Matched
Income, population, rents	Yes	Yes	Yes	Yes
Comparison	Means	Means	Means	Means
Observations	136	101	101	101
Countries	136	101	101	101
R ²	0.201	0.286	0.263	0.311

Notes: Each column is one regression of country means over 2003–2021. Tax rates are in percentage points and QoG is on its original 0–1 scale. QoG is lagged one year before averaging. TF denotes Tax Foundation. Columns (1)–(2) use each source’s available sample. Columns (3)–(4) use identical country-year observations before averaging. All columns control for log real GDP per capita, log population, and resource rents/GDP. Parentheses contain HC3 standard errors. ***, **, and * denote significance at 1, 5, and 10 percent. Appendix Table 14 reports all coefficients.

host. Residents’ outward direct investment is a separate series and is not deducted from inward FDI here.

For country h in year t , I estimate

$$\log(\text{Inward FDI}_{ht}) = \alpha_h + \lambda_t + \beta \text{QoG}_{h,t-1} + \delta' \text{Controls}_{ht} + u_{ht}. \quad (2)$$

The regression includes country and year fixed effects, with standard errors clustered by country. I use the annual income, population, and resource-rent variables from the tax comparison as controls. Columns (1) and (2) of Table 2 contain the same 902 observations in 56 countries over 2003–2019, so their difference reflects the inclusion of the controls.

Adding these controls lowers the QoG coefficient from 2.586 to 2.131. The latter implies that a 0.1 increase in the index is associated with 0.213 log points, approximately 24 percent more annual FDI. The country-clustered test gives $p = 0.071$, as does a wild-cluster bootstrap. No single country determines the sign of this association. Reestimating the regression after removing each country in turn always gives a positive coefficient. The inference and sample comparisons are reported in Appendix C.2.

2.3 Quality of Government and Profit Shifting

I use estimated net inward profit shifting as a percentage of GDP from Garcia-Bernardo and Jansky (2021). Positive values indicate net inward shifting, and negative values

TABLE 2: Quality of government and annual inward FDI

	(1)	(2)
Quality of government	2.586* (1.334)	2.131* (1.157)
Income, population, rents	No	Yes
Country effects	Yes	Yes
Year effects	Yes	Yes
Years	2003–2019	2003–2019
Observations	902	902
Countries	56	56
Within R^2	0.009	0.097

Notes: The dependent variable is log annual inward direct investment from Forbes–Warnock. Annual amounts sum all four quarters and are in current US dollar billions. Logs require positive annual flows. QoG is lagged one year and ranges from 0 to 1. Both columns include country and year effects and use identical observations. Parentheses contain country-clustered standard errors. ***, **, and * denote significance at 1, 5, and 10 percent. Appendix Table 15 reports all coefficients.

TABLE 3: Quality of government and estimated net inward profit shifting

	(1)	(2)
Quality of government	9.967*** (3.471)	9.995** (4.119)
Capital controls, growth, manufacturing	Yes	Yes
Statutory tax rate	Yes	Yes
Income and population	No	Yes
Year	2016	2016
Observations	116	116
Countries	116	116
R^2	0.177	0.179

Notes: Each column is one cross-sectional regression. Net inward shifting and manufacturing are percentages of GDP. Growth and tax rates are in percent. QoG ranges from 0 to 1. All explanatory variables refer to 2016. Both columns retain havens and negative net inward shifting. Parentheses contain HC3 standard errors, conditional on the published shifting estimates. ***, **, and * denote significance at 1, 5, and 10 percent. Appendix Table 16 reports all coefficients.

indicate net outward shifting. The 2016 cross section contains 116 countries, including 85 with negative values. I estimate

$$\text{Net shifting/GDP}_h = \alpha + \beta \text{QoG}_h + \delta' \text{Controls}_h + u_h. \quad (3)$$

I include capital controls on direct investment, GDP growth, manufacturing/GDP, and the statutory corporate tax rate in column (1), then add log real GDP per capita and log population in column (2). All explanatory variables refer to 2016. Both regressions use the same 116 countries and HC3 standard errors.

The estimated QoG coefficients in Table 3 are 9.967 and 9.995. In either specification, a 0.1 higher index is associated with approximately one percentage point of GDP more

net inward profit shifting. The coefficient in the second column has a p -value of 0.017, and it remains positive and significant at 5 percent after removing any single country. Appendix C.2 reports the sample checks.³

At the firm level, [Hopland et al. \(2018\)](#) document that payments within a multinational group adjust to operating losses in Norwegian tax records. The adjustment occurs within the tax year, largely in user fees such as royalties and licensing payments. In a study of transfer-pricing rules in 27 countries during 2006–2014, [de Mooij and Liu \(2020\)](#) find a relative decline in investment by multinational affiliates compared with domestic firms. These observations inform the timing of the model. A firm’s investment is already installed when it chooses its reported profits, but the return it expects from that later choice matters for investment.

2.4 Investment Following a Change in Treaty Protection

The decision in *Achmea* changed the enforcement of investment protection under intra-EU treaties. On March 6, 2018, the Court of Justice of the European Union ruled that the investor–state arbitration clause in an intra-EU bilateral investment treaty was incompatible with EU law ([Court of Justice of the European Union, 2018](#)). Firms covered by such treaties consequently faced a change in how they could enforce protection against their host government. I examine their investment around the judgment.

The investment data come from Eurostat’s inward foreign affiliates statistics (FATS) ([Eurostat, 2026](#)). They report gross investment in tangible goods by foreign-controlled enterprises, in millions of euros, for each host and ultimate-parent country pair over 2010–2019. I identify treaty coverage just before *Achmea* from the lists attached to the termination agreement ([European Union, 2020](#)). For inclusion, both the host and the parent country must be parties to that agreement. I drop pairs whose treaties had already been terminated. There are 22 host and 22 parent countries in the resulting panel.

I estimate the annual investment coefficients by Poisson pseudo-maximum likelihood (PPML), specifying the conditional mean as an exponential function. Investment enters in levels, so a reported zero can be used without a log transformation. The regressions allow a separate fixed effect for each country pair and for each host-year and parent-year combination. These absorb permanent differences across pairs, together with annual tax and macroeconomic conditions common to firms operating in a host or headquartered in a parent country. In the second regression I allow investment within

³Net inward shifting/GDP aggregates transfers across jurisdictions. It differs from an individual firm’s outward transfer divided by its host-country profits, both in coverage and in the denominator.

TABLE 4: Investment after the Achmea judgment

	(1)	(2)
Post-judgment coefficient	−0.474** (0.215)	−0.648** (0.280)
Post effects	By year	By year
Pair-specific trends	No	Yes
Pre-period joint p -value	0.017	0.549
Observations	2,622	2,547
Host countries	22	22
Parent countries	22	22
Country pairs	349	320

Notes: Each column reports one annual event-study PPML regression for 2010–2019. The post-judgment coefficient averages the 2018 and 2019 coefficients. Both regressions include pair, host-year, and parent-year effects and use 2017 as the reference year. Column (2) also includes pair trends and normalizes 2010 to zero. Parentheses contain standard errors clustered by host and parent country, using 21 reference degrees of freedom. The joint pre-period tests use host-clustered covariance matrices. ***, **, and * indicate significance at 1, 5, and 10 percent.

each pair to follow its own linear trend as well. I use standard errors clustered by host and parent. The equation and all annual coefficients appear in Appendix C.4.

Averaging the two post-judgment coefficients gives -0.474 without trends and -0.648 with trends (Table 4). The latter estimate has a standard error of 0.280. Thus the investment decline appears under either specification. Before the judgment, the annual coefficients reject joint equality to zero in the no-trend regression. With trends, the joint test instead gives $p = 0.549$. The second estimate is interpreted relative to the investment path implied by continuing each pair’s fitted trend.⁴

The treaty comparison concerns enforcement of investment protection. In the model, new firms respond to the information conveyed by the government’s tax decision. Their investment gives the government future revenue to lose if it raises the tax on existing firms. The reporting opportunities of both cohorts enter the comparison between maintaining policy and revising it.

3 Model

I consider corporate taxation with investment preceding the implemented tax and profit shifting following it. Commitment requires the government to implement its initial policy. Without commitment, it can reoptimize on installed capital, accounting for the firm’s ability to adjust reported profits. I solve these decisions in reverse order.

⁴The panel has 352 observed pairs in 2017, 99 in 2018, and 334 in 2019. Restricting it to pairs observed in all ten years would leave only six host countries. I use the available observations and retain the full-coverage comparison in the analysis files. Exposure is based on ultimate-parent nationality, which may not be the nationality used to claim treaty protection through an intermediate holding company.

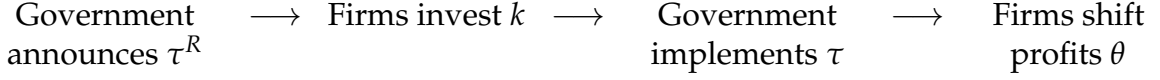


FIGURE 1: Timing within a period

3.1 Environment and Timing

There is a unit measure of identical, foreign-owned multinational firms. The host government applies one common tax rate, with no firm-specific taxation. Each firm allocates a fixed capital endowment $\bar{k} > 0$ between the host country and a foreign location. Investment k in the host produces profits $\rho(k)$, while investment $\bar{k} - k$ abroad produces $\rho(\bar{k} - k)$. Profits are net of the rental cost of capital. The profit function satisfies $\rho(0) = 0$ and is increasing and strictly concave on the investment domain. The parameterized comparisons use $\rho(k) = zk^\alpha - r^*k$, where $z > 0$, $0 < \alpha < 1$, and $r^* \geq 0$ is the rental rate. The host is a small open economy that takes the foreign corporate tax rate $\tau^* \in (0, 1)$ as given.

There are two government types, both of which maximize host corporate tax revenue. The commitment type must implement the tax announced before investment. The opportunistic type can reoptimize after firms invest, taking advantage of the irreversibility of their investment. Firms initially do not observe type and assign probability p to commitment. Following the interpretation in Section 2, I call this prior government reputation.

Write τ^R for the tax policy chosen before investment and implemented by the commitment type. The realized host rate is $\tau \in [0, 1]$. Firms choose their profit-shifting amount θ after observing τ . A positive θ moves profits into the host country and a negative θ moves profits out. All announcements and implemented rates are public. The choices occur in the following order.

Capital is fixed for the remainder of the period once it has been invested. Intragroup payments can still be changed within the firm's existing structure before taxes are filed, consistent with the evidence in Section 2.3. I begin with that reporting decision.

3.2 Stage 4. Firms' Profit-Shifting Choice

Given its capital and both tax rates, the firm chooses θ to maximize its total after-tax profit,

$$\begin{aligned} \max_{\theta} \quad & (1 - \tau) \left[\rho(k) + \theta - \frac{\gamma\theta^2}{2\rho(k)} \right] \\ & + (1 - \tau^*) \left[\rho(\bar{k} - k) - \theta - \frac{\gamma\theta^2}{2\rho(\bar{k} - k)} \right], \\ \text{subject to} \quad & \rho(k) + \theta - \frac{\gamma\theta^2}{2\rho(k)} \geq 0, \\ & \rho(\bar{k} - k) - \theta - \frac{\gamma\theta^2}{2\rho(\bar{k} - k)} \geq 0. \end{aligned} \quad (4)$$

The resources used in arranging transfers, such as legal and accounting services, are governed by $\gamma > 0$. Costs are deductible in each country. I require both reported tax bases to be nonnegative, including when the host taxes profits at a rate of one.

Let $\theta(k, \tau)$ denote the firm's optimal transfer. When neither constraint binds,

$$\theta(k, \tau) = \frac{(\tau^* - \tau)\rho(k)\rho(\bar{k} - k)}{\gamma[(1 - \tau)\rho(\bar{k} - k) + (1 - \tau^*)\rho(k)]}. \quad (5)$$

The tax difference determines the direction of the transfer in (5). For $\tau > \tau^*$, the firm reports less profit in the host country as the host rate rises. This adjustment continues until it would make one of the tax bases negative, at which point the firm chooses the nearest feasible transfer. The amount that can be shifted tends to zero as investment in either country tends to zero. I derive the solution with these constraints in Appendix A.1.

3.3 Stage 3. Optimal Taxation under Discretion

At this stage, aggregate host capital K is already installed. Write $\Lambda(K)$ for aggregate profits before shifting and $\Theta(K, \tau)$ for aggregate profit shifting. The unit measure of identical firms gives $\Lambda(K) = \rho(K)$ and $\Theta(K, \tau) = \theta(K, \tau)$ in a symmetric allocation. The commitment type implements τ^R . The opportunistic type chooses the tax that maximizes current revenue,

$$\tau^O(K) \in \arg \max_{\tau \in [0,1]} \tau \left[\Lambda(K) + \Theta(K, \tau) - \frac{\gamma\Theta(K, \tau)^2}{2\Lambda(K)} \right]. \quad (6)$$

The opportunistic government trades off the tax rate against the reported base, taking installed capital as given. Unlike a government taxing immobile profits, it may collect more at a rate below one because firms move profits abroad in response to higher taxes.

3.4 Stage 2. Firms' Investment Choice

Firms invest after observing the announcement and before learning the implemented tax. When both government types announce the same rate, the announcement conveys no information about type. Firms then expect τ^R with probability p and $\tau^O(K)$ with probability $1 - p$. Each firm chooses capital to maximize

$$\max_{k \in [0, \bar{k}]} \mathbb{E}_\tau \left\{ (1 - \tau) \left[\rho(k) + \theta(k, \tau) - \frac{\gamma \theta(k, \tau)^2}{2\rho(k)} \right] + (1 - \tau^*) \left[\rho(\bar{k} - k) - \theta(k, \tau) - \frac{\gamma \theta(k, \tau)^2}{2\rho(\bar{k} - k)} \right] \right\}. \quad (7)$$

The expectation uses those two implemented taxes. An atomistic firm takes them as given and does not internalize the effect of aggregate capital on the government's tax choice. Symmetry requires its chosen k to equal K . This condition and (6) jointly characterize investment and discretionary taxation. I write $K(\tau^R, p)$ for capital in the equilibrium under consideration.

Profit shifting affects investment because it changes the return firms retain after taxation. Better reputation puts more weight on the rate under commitment in this comparison. A tax reduction can consequently attract more capital when firms consider it more likely to be implemented.

3.5 Stage 1. Optimal Taxation under Commitment

Both government types choose an announcement anticipating investment and the subsequent tax and profit-shifting decisions. I study pooling equilibria in which they announce the same rate. Following an unexpected announcement, firms believe the government is opportunistic with certainty. They then choose the investment appropriate for a known opportunistic government. Neither type can obtain more from such an announcement than the maximum tax revenue on that investment. A common announcement must give each type at least this amount.

For a given announcement, tax revenue at reputation p and implemented rate τ is

$$R(p, \tau) = \tau \left[\Lambda(K) + \Theta(K, \tau) - \frac{\gamma \Theta(K, \tau)^2}{2\Lambda(K)} \right], \quad K = K(\tau^R, p).$$

In this notation, τ^R determines the investment on which revenue is collected. At $p = 0$, firms expect the discretionary tax regardless of the announcement. I fix the same known-type investment equilibrium after every unexpected announcement. Thus $\max_{\tau \in [0,1]} R(0, \tau)$ is the largest revenue either type can obtain after an unexpected announcement.

The commitment type's preferred implementable announcement solves

$$\begin{aligned} \max_{\tau^R \in [0,1]} \quad & \tau^R \left[\Lambda(K) + \Theta(K, \tau^R) - \frac{\gamma \Theta(K, \tau^R)^2}{2\Lambda(K)} \right] \quad K = K(\tau^R, p), \\ \text{subject to} \quad & R(p, \tau^R) \geq \max_{\tau \in [0,1]} R(0, \tau), \\ & R(p, \tau^O(K(\tau^R, p))) \geq \max_{\tau \in [0,1]} R(0, \tau). \end{aligned} \quad (8)$$

The first inequality prevents the commitment type from changing its announcement. The second prevents the opportunistic type from doing so, accounting for its optimal tax revision after investment. They make the common announcement implementable. They do not require the opportunistic type to honor the announced tax, since its Stage 3 choice is still given by (6).

The solution is the optimal tax under commitment at reputation p . It maximizes the commitment type's revenue, accounting for the effect of its tax on investment and profit shifting and for firms' uncertainty about type. The maximization selects among supported pooling equilibria. A unilateral change in the announcement would instead trigger the off-path beliefs specified above. Appendix A.2 verifies the equilibrium and both announcement incentives.

When $p < 1$, commitment is private information and firms allow for a discretionary revision. The optimal tax under full commitment is the special case $p = 1$, when commitment is common knowledge. The government maximizes corporate tax revenue. A national welfare analysis would additionally require domestic factor incomes, ownership, and public expenditure.

4 Optimal Taxation under Commitment and Discretion

This section studies how reputation changes optimal taxation, holding production, the foreign tax, and profit-shifting opportunities fixed. The commitment type chooses its policy taking account of investment. The opportunistic type chooses its tax after capital is installed. I first compare these choices with and without profit shifting, then establish when a limit on transfers makes confiscation optimal under discretion. The numerical

analysis examines the bilateral cost model and its sensitivity to the cost specification and foreign tax.

4.1 Optimal Taxation with Profit Shifting

An improvement in reputation has two countervailing effects on the commitment type's preferred rate. If $\tau^R < \tau^O$, it increases investment at a given τ^R , allowing the government to collect revenue from a larger base. It also increases the distortion from a marginal tax increase, because firms put more weight on τ^R in making their investment decisions. The second effect favors a lower rate.

Profit shifting changes the relative strength of these effects. Firms can retain part of their investment return even under an opportunistic government, so an improvement in reputation need not generate as large an expansion of investment as it would without shifting. If firms have enough protection against discretionary taxation, the investment response to the announced rate determines the comparison. The commitment type then chooses a lower rate as p rises. To establish this result analytically, let profits be $\rho(k) = zk^\alpha - r^*k$, with $z > 0$, $0 < \alpha < 1$, and $r^* \geq 0$, and let shifting costs be deductible only in the country from which profits are transferred. The host deducts shifting costs only on an outward transfer. With inward shifting its tax base is $\rho(K) + \Theta(K, \tau)$, and the cost is deducted abroad. Tax bases remain nonnegative. The timing, government types, and pooling selection are unchanged.

Suppose the parameters satisfy

$$\begin{aligned} \alpha z \bar{k}^{\alpha-1} &> (2 - \alpha)r^*, \\ \gamma &> \frac{\tau^*}{1 - \tau^*} + \frac{(\tau^*)^2}{2(1 - \tau^*)^2}, \\ \frac{\tau^*}{\gamma(1 - \tau^*)} \left(\frac{\sqrt{1 + 2\gamma} + 1}{2} \right)^{\alpha/(1-\alpha)} + \frac{\tau^* \alpha}{(1 - \alpha)(1 - \tau^*)} &< 1, \\ \frac{\sqrt{1 + 2\gamma} - 1}{2} \left[\frac{1}{2} + \frac{(2 - \alpha)r^*}{\alpha z (\bar{k}/2)^{\alpha-1}} \right] &< 1 - \alpha. \end{aligned} \tag{9}$$

The first condition ensures positive marginal profits and a declining elasticity of profits with respect to retained returns. The next two place the relevant taxes above the foreign rate and below the rate that exhausts the host tax base. The last ensures that profit shifting protects enough of the investment return under discretion. Appendices A.3 and A.4 derive the investment response and the retained-return bound.

Proposition 1 (Lower optimal taxes with profit shifting). *Consider origin-country deductible shifting costs and $\rho(k) = zk^\alpha - r^*k$, with $z > 0$, $0 < \alpha < 1$, $r^* \geq 0$, and the four parameter restrictions in (9). Let $\tau^R(p)$ be any optimal tax under commitment selected as in Section 3.5. Then $\tau^* < \tau^R(p) < \tau^O$ for every $p \in (0, 1]$, and, for every $0 < p < p' \leq 1$,*

$$\tau^R(p') < \tau^R(p), \quad K(\tau^R(p'), p') > K(\tau^R(p), p).$$

These inequalities hold for every choice of maximizer at the two reputation levels.

Proof. See Appendix A.5. □

Proposition 1 places the optimal tax between the foreign rate and the discretionary rate. At this policy, an increase in reputation makes a reduction in τ^R more effective in attracting capital. The conditions in (9) ensure that this investment response is strong enough to induce the government to reduce its rate. Firms invest more and shift a smaller fraction of their host profits abroad, expanding the reported base sufficiently for tax revenue to rise as well.

Two further implications follow. First, the opportunistic type also earns more revenue at higher reputation because it inherits a larger investment base. Second, the expected implemented tax and expected outward profit shifting as a share of host profits both fall. The tax under commitment is lower and is implemented more often. Both types earn at least the revenue available after an unexpected announcement, so their announcement incentives also hold.

The selection of the tax under commitment can also be disciplined by announcement incentives across equilibria. Under the equilibrium criterion of Lu (2013), a proposed deviation is assessed using the types that would prefer an alternative equilibrium with that announcement. In this benchmark, the surviving pure announcement outcomes are exactly those preferred by the commitment type. Appendix A.6 establishes this result and compares pooling with separating announcements. The comparison in Proposition 1 consequently applies to every tax that survives this criterion.⁵

The next comparison removes profit shifting while retaining the same production technology. It isolates the protection against a tax revision that drives Proposition 1.

⁵Under the same assumptions and $0 < p < 1$, another pooling equilibrium has both types announce and implement τ^O , with belief zero after any other announcement. Investment is $K(\tau^O, 0)$, and both types receive $R(0, \tau^O)$. This equilibrium survives the intuitive criterion of Cho and Kreps (1987). At any investment response, a deviation profitable for the commitment type is also profitable for the opportunistic type, which can implement the same tax and has the same equilibrium payoff. Belief zero therefore remains admissible. Tax and investment are independent of p in this equilibrium, so the reputation comparison depends on equilibrium selection.

4.2 Reputation without Profit Shifting

Without profit shifting, installed capital and the profits it generates are both exposed to host taxation. The opportunistic government chooses $\tau^O = 1$. Firms receive host profits only when the government has commitment, so investment satisfies

$$p(1 - \tau^R)\rho'(K) = (1 - \tau^*)\rho'(\bar{k} - K). \quad (10)$$

Better reputation now protects returns that would otherwise be confiscated. Its effect on investment is sufficiently strong that the government chooses a higher tax under commitment.

Proposition 2 (Higher optimal taxes without profit shifting). *Suppose profit shifting is unavailable and $\rho(k) = zk^\alpha - r^*k$, with $z > 0$, $0 < \alpha < 1$, $r^* \geq 0$, and $\alpha z \bar{k}^{\alpha-1} > (2 - \alpha)r^*$. For every $p \in (0, 1]$, (8) has a unique maximizer $\tau^R(p) \in (0, 1)$. For every $0 < p < p' \leq 1$,*

$$\tau^R(p') > \tau^R(p), \quad K(\tau^R(p'), p') > K(\tau^R(p), p).$$

Proof. See Appendix A.7. □

Profit shifting reverses the tax comparison by changing investment under the opportunistic type. Without shifting, firms lose the entire host return when the government is opportunistic. An improvement in reputation then has a sufficiently large effect on investment that the commitment type prefers to raise its rate. With shifting, firms obtain part of the return even after a tax revision, and some investment would take place without commitment. Under the conditions of Proposition 1, the government instead lowers its rate to exploit the greater responsiveness of investment to τ^R .

This comparison concerns the choice of τ^R . The expected implemented rate without shifting, $p\tau^R + (1 - p)$, still falls with p as confiscation becomes less likely. An expectation over the two government types is distinct from either the chosen policy or a realized statutory rate.

4.3 When Profit Shifting Restrains Taxation

The difference between the preceding results depends on how much profit firms can move after a large tax increase. Their response to a small tax difference need not reveal that amount. To examine this distinction, maintain the origin-country benchmark and the parameter conditions of Proposition 1. Take installed capital from any equilibrium allocation in that proposition and hold it fixed while limiting outward transfers to a fraction of host profits. The comparison changes the firm's ability to move profits after investment.

The unrestricted discretionary tax τ^O is the unique root above τ^* of

$$(2\gamma + 1)(1 - \tau^O)^3 = (1 - \tau^*)^2(1 + \tau^O).$$

The permitted outward share lies in $[0, (\sqrt{1 + 2\gamma} - 1)/\gamma]$. The critical share makes the government indifferent between the unrestricted tax and confiscation. It equals

$$\frac{\sqrt{1 + 2\gamma \left\{ 1 - \tau^O \left[1 + \frac{\Theta(K, \tau^O)}{\Lambda(K)} - \frac{\gamma \Theta(K, \tau^O)^2}{2\Lambda(K)^2} \right] \right\}} - 1}{\gamma}, \quad (11)$$

where $\Theta(K, \tau^O)$ is the transfer without the additional limit.

Proposition 3 (Transfer limits and confiscatory taxation). *Maintain the assumptions of Proposition 1 and fix $K = K(\tau^R(p), p)$ at any $p \in (0, 1]$. For the admissible transfer limits just defined, the unique revenue-maximizing tax is 1 when the permitted share is below (11) and τ^O when it is above. At equality, exactly these two taxes maximize revenue.*

Proof. See Appendix A.8. □

The reason for the result is that a binding limit stops the firm's outward transfer from increasing with the tax. The government can raise its rate on the remaining base without losing more profits abroad. If the permitted transfer is small enough, taxing that base at one yields more than the unrestricted optimum. The threshold in (11) equates revenue at these two rates.

The critical limit is greater than the actual transfer at τ^O . Consequently, the government can prefer confiscation even though firms' transfers at the old tax have not changed. A positive limit also leaves the reporting response near equal tax rates unchanged, since it does not bind in a sufficiently small neighborhood. That local response thus omits an adjustment relevant to a large tax increase. Capital is held fixed in the proposition. The numerical example compares the transfer needed to deter confiscation with the transfer made at the unrestricted optimum.

Avoidance restrictions affect the behavioral response relevant for tax-system design (Slemrod and Kopczuk, 2002). The comparison here concerns the government's incentive to revise its tax after investment. It requires the reporting response over the range of possible tax choices, since a restriction can alter the most profitable revision without changing the tax base near the existing rate.

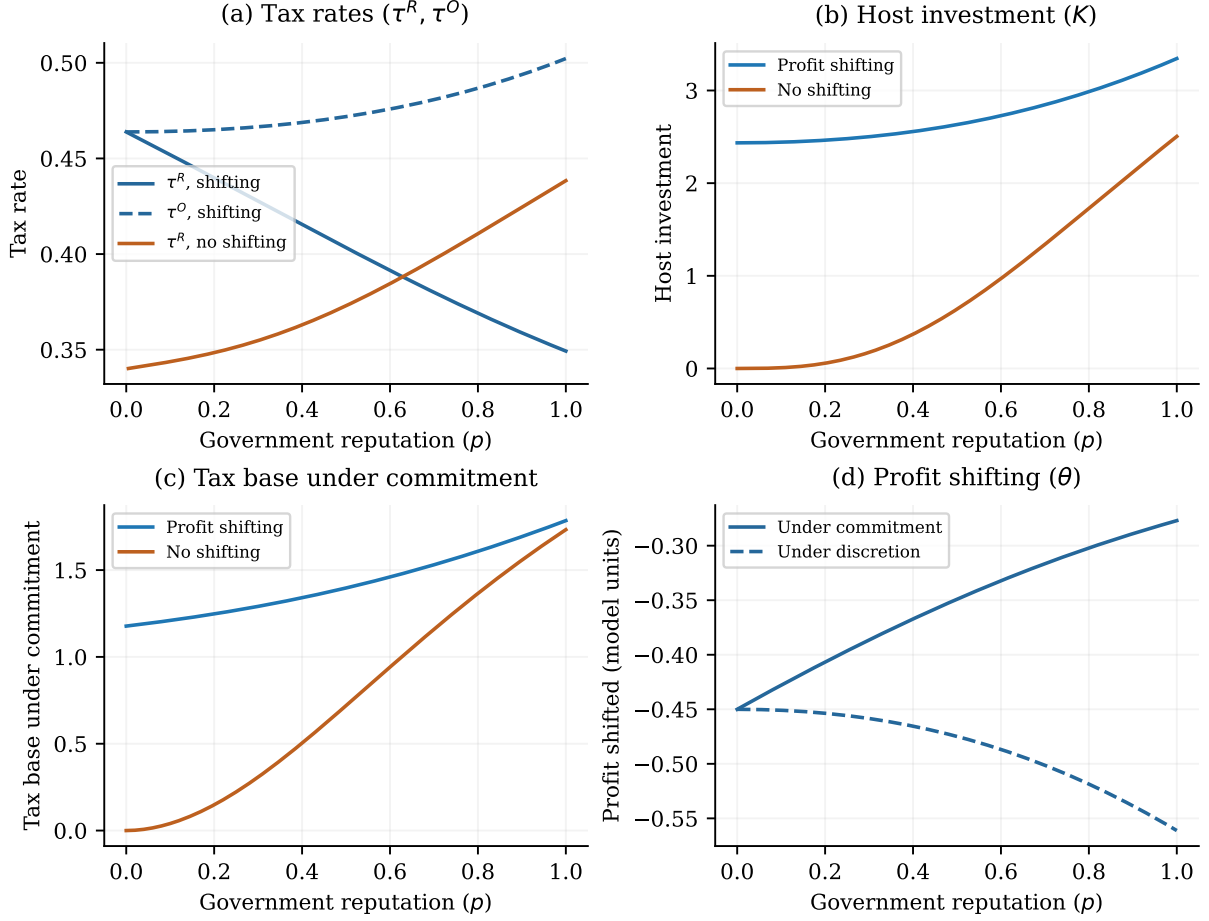


FIGURE 2: Reputation, tax policy, investment, and profit shifting

Notes: All panels use the baseline production parameters stated in the text. Shifting costs are deductible in both countries. Panel (a) distinguishes taxation under commitment from taxation under discretion. Panel (c) reports the tax base under commitment. Panel (d) reports profit shifting in model units, with negative values denoting outward shifting. The opportunistic-type tax without shifting is one.

4.4 Numerical Illustration

The tax response affects both where firms invest and where they report profits. I compute these joint choices in the baseline with deductible costs in both countries. I set $z = 1$, $\alpha = 0.66$, $r^* = 0.04$, $\bar{k} = 10$, $\tau^* = 0.15$, and $\gamma = 1.25$ in $\rho(k) = zk^\alpha - r^*k$, holding these parameters fixed as reputation changes. The comparison without shifting uses the same production parameters. The parameter values illustrate the mechanism. Appendix B.6 describes the solution and verification procedures.

Figure 2 gives the same declining commitment-tax response as the analytical model. Increasing p from 0.1 to 0.9 lowers τ^R from 45.2 to 35.9 percent and raises K from 2.442 to 3.153. At the commitment rate, the reported base expands from 1.210 to 1.693, while the outward transfer falls from 0.428 to 0.289. Hence revenue increases from 0.547 to 0.607 despite the reduction in the rate.

Under the opportunistic type, the extra capital is already in place when the tax is chosen. Its optimal rate rises from 46.4 to 49.4 percent, and firms respond by raising their outward transfer from 0.451 to 0.538. Thus higher reputation leads to less outward shifting if the government has commitment and more if it is opportunistic. This dependence of τ^O on capital comes from the costs incurred in both countries. Under origin-country costs, τ^O is constant.

Expected shifting averages the transfers across the two types. In the origin-country model, the expected outward share of host profits falls with p . The corresponding amount also reflects the increase in host profits as investment expands.

The numerical example also shows how observed transfers can understate the capacity needed to restrain taxation. At $\gamma = 1.25$ and $\tau^* = 0.15$ in the origin-country model, firms transfer 24.2 percent of host profits at the unrestricted opportunistic tax. They must be able to transfer more than 55.6 percent for the government to prefer that rate to a tax of one. The first share is observed under the lower tax. The second is the capacity that deters confiscation, as derived in Proposition 3.

I repeat the tax comparison with non-deductible costs and with costs at the origin alone. I choose cost parameters to give the same reporting response at equal taxes and equal profits. The reference semi-elasticities are 1.26 and 1.50, guided by [Kallen \(2023\)](#) and [Beer et al. \(2018\)](#). All six comparisons give a tax under commitment that declines with reputation. Appendix B.3 sets out the cost functions and reports the matched parameters and taxes.

4.5 Sensitivity to Costs and Foreign Taxes

The preceding results depend on how much protection profit shifting provides after a tax increase. I now vary the level and curvature of shifting costs, allow costs to differ across countries, and change the foreign tax rate. Each comparison retains nonnegative tax bases and both types' announcement incentives. Appendix B.1 varies production curvature, and Appendix B.2 gives the alternative cost functions and numerical allocations.

With quadratic costs, higher costs allow the government to tax more heavily after investment. The tax under commitment is also higher and investment is lower. For $\gamma = 0.5, 1.25, 1.5$, the tax under commitment nevertheless falls at each of the five reputation levels in panel (a). The same ordering holds when only the foreign cost changes, with $\gamma^* = 0.5, 1.25, 2$ and the home cost fixed. At $p = 0.5$, raising the foreign cost over this range raises the tax under commitment from 36.5 to 43.0 percent and lowers host investment from 3.139 to 2.266. Thus the declining tax under commitment does not require equal costs in the two countries.

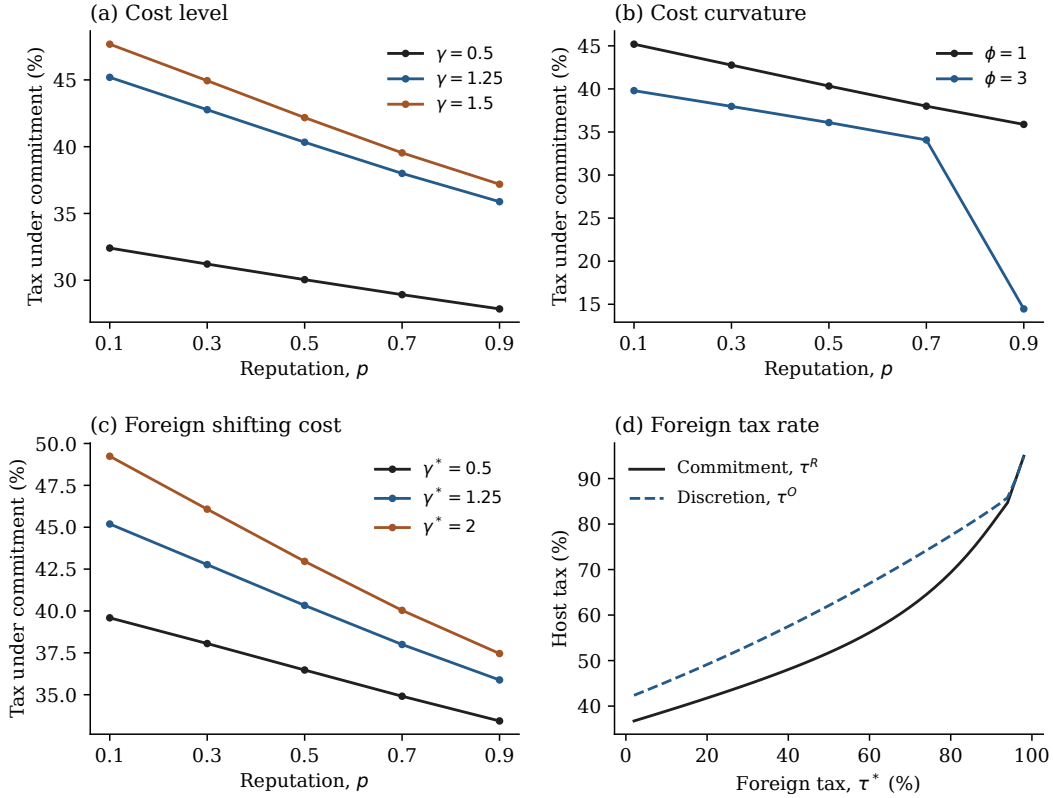


FIGURE 3: Tax policies under alternative costs and foreign taxes

Notes: Panels (a)–(c) vary reputation at fixed parameters. Panel (a) changes the common quadratic cost parameter. Panel (b) compares quadratic and quartic costs. In each country, quartic costs equal $\gamma\theta^4$ divided by twice the cube of profits before shifting. Panel (c) fixes the home cost at $\gamma = 1.25$ and varies the foreign cost γ^* . Panel (d) varies the foreign tax at $p = 0.5$. Other parameters equal those in Figure 2.

Changing the power of the cost function can also change where profits are reported. Under the quartic specification, the tax under commitment falls from 39.8 percent at $p = 0.1$ to 14.5 percent at $p = 0.9$. At the higher reputation it lies below the foreign rate, so the host attracts shifted profits. The declining tax response therefore also appears when high-reputation governments attract reported profits from abroad. With sixth-power costs and a foreign tax of 30 percent, the tax under commitment moves only from 28.7 to 28.4 percent over the same reputation comparison. Reputation then has little effect on the tax under commitment. These changes in the cost function affect both its level and curvature, rather than holding the local reporting elasticity fixed.

Panel (d) varies the exogenous foreign tax from 2 to 98 percent at $p = 0.5$. The host tax under commitment and the discretionary tax both increase, and more capital is allocated to the host. At the upper foreign rates, firms also report profits in the host that were earned abroad. This exercise describes the host's adjustment to foreign taxation, with no feedback to the foreign rate.

To examine heterogeneity, I use two groups of firms with different shifting costs. This is motivated by evidence that shifting is concentrated in large firms and that firms

incur different costs of tax planning (Wier and Erasmus, 2023; Bilicka et al., 2024). Each group invests separately, but the government applies one rate to their combined base. All nine comparisons of costs and population shares give a tax under commitment that falls with reputation. Appendix B.4 describes these numerical exercises. If separate rates can be set for observable groups, the same result follows analytically whenever each group's parameters satisfy Proposition 1. The proof in Appendix A.11 verifies the announcement constraints for the policy formed from the group optima.

5 Reputation and Sustainable Taxation

The previous section considers a government that has no further cohort to attract after choosing its implemented tax. I now add a second cohort. The one-period equilibrium describes its behavior after observing either implementation or a revision, allowing the government's immediate revenue gain to be compared with its subsequent loss.

5.1 A Two-Period Model

The horizon has two periods. A unit measure of firms enters in each period with capital $\bar{k}$ and the production and reporting opportunities specified in Section 3. Profits accrue in the period of entry only, with no capital carried to the next cohort. Government type is fixed over the two periods and known to the government, which discounts its period-2 revenue by $\beta \in (0, 1)$. The initial prior about commitment is $p \in (0, 1)$.

Period 1 follows the timing of the one-period model. After the announcement τ_1^R , firms allocate capital and the government chooses the implemented tax. Firms then make their profit transfers. The commitment type implements τ_1^R by assumption. For the opportunistic type, a different tax in $[0, 1]$ remains feasible, but the decision must now account for the information conveyed to the next cohort.

Firms entering in period 2 observe the first-period announcement and implementation before hearing the new announcement, τ_2^R . They choose investment given this history. The commitment type implements its rate, and the opportunistic type reoptimizes exactly as in the one-period model since no further investment follows. Profit shifting is chosen last. The first cohort's experience thus informs the second cohort's investment through beliefs about the government's type.

I consider a common first-period announcement that both types implement. This behavior leaves the prior unchanged at p . If the government instead revises the tax, firms learn that it is opportunistic and the probability of commitment becomes zero. Continuation play at either belief is selected as in Section 3.5. I denote the opportunistic type's second-period tax by τ_2^{RO} after maintaining its policy and τ_2^{OO} after reneging. In

the bilateral-cost model, different investments under these beliefs can imply different discretionary rates.

Equilibrium requires optimal investment and profit transfers by each cohort and no profitable tax revision in period 1. Both types must also prefer the common announcement to any other announcement. After an unexpected announcement, firms believe they face the opportunistic type. Within the equilibria supported by these beliefs, I choose the announcement that gives the commitment type the highest revenue.

5.2 The Incentive to Maintain the First-Period Tax

Consider a first-period tax that both types implement. Firms face a certain tax τ_1^R , even though they remain uncertain about the government type. Their investment solves (7) with that tax imposed with probability one, giving $K(\tau_1^R, 1)$. At a fixed first-period tax, this investment does not depend on p . Reputation matters for the government's future revenue rather than the current return firms expect under that policy.

Extend the revenue notation from Section 3.5 to an implementation probability π . For the bilateral cost specification, revenue in either period is

$$R(\pi, \tau) = \tau \left[\Lambda(K) + \Theta(K, \tau) - \frac{\gamma \Theta(K, \tau)^2}{2\Lambda(K)} \right], \quad K = K(\tau^R, \pi). \quad (12)$$

Here π is the probability of implementing the announcement when firms invest, and τ is the tax subsequently imposed. The announcement is understood when writing $R(\pi, \tau)$. Thus $R(1, \tau_1^R)$ and $R(1, \tau_1^O)$ use the same installed first-period capital. The tax τ_1^O is the most profitable revision at that capital. In the final period, the corresponding implementation probabilities are p after honoring and zero after reneging. With origin-country costs, the host deducts the shifting cost only for an outward transfer. For inward shifting, its revenue is $\tau[\Lambda(K) + \Theta(K, \tau)]$.

The opportunistic government can honor its tax policy and retain its reputation, or it can collect more revenue now and face firms that understand its type in period 2. It prefers to honor the commitment when

$$\underbrace{R(1, \tau_1^O) - R(1, \tau_1^R)}_{\text{current revenue gain from reneging}} \leq \beta \underbrace{\left[R(p, \tau_2^{RO}) - R(0, \tau_2^{OO}) \right]}_{\text{future revenue lost after reneging}}. \quad (13)$$

The left side accounts for the profit shifting induced by a tax increase on already installed capital. The right side accounts for the next cohort's investment response to learning the government's type. Section 4 supplies both continuation outcomes. In

particular, Proposition 1 implies that the future revenue loss increases with p in the analytical benchmark.

The second cohort invests $K(\tau^O, 0)$ if it has observed reneging in the origin-country benchmark. Following implementation, it invests $K(\tau^R(p), p)$ instead, which is larger by Proposition 1. This difference determines the loss of future revenue. For the first cohort, a higher implemented tax raises outward transfers but leaves capital fixed. The two margins of adjustment are therefore reporting by existing firms and investment by new firms.

Before firms invest, each government type can choose a different announcement. At the specified off-path beliefs, its revenue from doing so cannot exceed the discounted revenue obtained when the opportunistic type is known. Implementation must therefore be accompanied by the following announcement constraints

$$\begin{aligned} R(1, \tau_1^R) + \beta R(p, \tau_2^R) &\geq (1 + \beta)R(0, \tau_2^{OO}), \\ R(1, \tau_1^R) + \beta R(p, \tau_2^{RO}) &\geq (1 + \beta)R(0, \tau_2^{OO}). \end{aligned} \quad (14)$$

The first inequality concerns the commitment type and the second the opportunistic type. Because honoring preserves the same belief p at every candidate first-period tax, continuation revenue is independent of the first-period rate. The commitment type's preferred announcement thus maximizes $R(1, \tau_1^R)$ subject to (13) and (14). This is the tax choice studied next.

Profit-shifting opportunities affect both sides of (13). They restrain the revenue obtainable from the installed base, but also preserve investment after the government's type becomes known. The latter can reduce the future cost of reneging. To isolate the effect of reputation, I hold profit-shifting opportunities fixed when comparing sustainable taxes.

5.3 Optimal Sustainable Taxation

With full commitment, the government chooses the tax that maximizes revenue after firms have invested in anticipation of that rate. Without commitment, a low tax can be difficult to sustain. It attracts capital but collects less tax on the installed base, increasing the gain from a subsequent tax rise. Condition (13) determines how far the government can lower the tax while retaining the opportunistic type's incentive to honor it.

The next result uses the origin-country cost specification and the same parameter restrictions as Proposition 1. In this specification, the optimal discretionary tax is independent of installed capital, so $\tau_1^O = \tau_2^{RO} = \tau_2^{OO} = \tau^O$. Write $\tau^R(1)$ for the optimal tax under full commitment. The continuation equilibrium is selected as in Section 3.5.

Proposition 4 (Lower taxes sustained by reputation). *Suppose shifting costs are deductible in the country of origin, $\rho(k) = zk^\alpha - r^*k$ with $z > 0$, $0 < \alpha < 1$, and $r^* \geq 0$, and the four restrictions in (9) hold. Fix $\beta \in (0, 1)$. Among the pooling equilibria in Section 5 in which both types honor, the optimal first-period tax $\tau_1^R(p)$ is unique. It equals $\tau^R(1)$ if that tax satisfies (13). Otherwise, it is the unique tax in $(\tau^R(1), \tau^O)$ at which (13) binds. For every $0 < p < p' < 1$,*

$$\tau_1^R(p') \leq \tau_1^R(p), \quad \text{strictly if } \tau_1^R(p) > \tau^R(1).$$

Both announcement constraints in (14) hold at this tax.

Proof. See Appendix A.9. □

To understand Proposition 4, start where the implementation constraint binds. Increasing p gives the government more second-period investment and revenue if it maintains policy. Its revenue after revealing the opportunistic type is unchanged. The government therefore has more to lose by reneging and will accept a larger sacrifice of revenue on the installed first-period capital. Anticipating this willingness, firms invest more at a lower initial tax. The commitment type chooses that lower rate because it raises current revenue.

The rate falls until it reaches $\tau^R(1)$. At that point, another reduction would lower revenue even with certain implementation, so the government makes no further cut. Increasing β has the same effect as increasing the value of retained reputation, since it places more weight on the revenue from future cohorts of firms.

In this equilibrium, firms in period 1 know that their tax will be implemented. At a given τ_1^R , their investment is $K(\tau_1^R, 1)$ regardless of p . It is the lower chosen tax that raises their investment as reputation improves. The role of reputation in sustaining that tax comes from the next cohort, whose investment would fall after a revision.

The result extends to every finite horizon with a new cohort in each period. If both types maintain policy before the final period, higher reputation increases the continuation revenue at risk at every earlier date. Backward induction therefore gives a weakly lower optimal tax in each of those periods, with a strict decline until the full-commitment tax is reached. Appendix A.10 states the extended environment, writes each implementation constraint, and proves the result.⁶

⁶With infinitely many new cohorts and no profit shifting, suppose a tax revision leads all subsequent cohorts to expect confiscation and invest zero. If both types otherwise implement the same $\tau^R < 1$ forever, investment is $K = K(\tau^R, 1)$. For positive host profits, the implementation constraint is $(1 - \tau^R)\rho(K) \leq \beta\tau^R\rho(K)/(1 - \beta)$, or $\tau^R \geq 1 - \beta$. Future tax revenue can therefore sustain positive investment even when the government is known to be opportunistic. The initial reputation does not enter this stationary constraint. The finite-horizon reputation effect instead operates through the selected terminal continuation.

TABLE 5: Reputation and the tax that both government types honor

Shifting cost γ	Reputation p	Tax (%)	Investment	Current revenue
0.50	0.1	32.5	3.589	0.499
0.50	0.5	30.1	3.775	0.514
0.50	0.9	27.6	3.980	0.520
1.25	0.1	45.4	2.510	0.559
1.25	0.5	40.9	2.854	0.602
1.25	0.9	35.8	3.272	0.623
4.00	0.1	60.0	1.336	0.535
4.00	0.5	52.0	1.884	0.646
4.00	0.9	42.8	2.629	0.706

Notes: The tax is the first-period announcement, implemented by both types. Investment and current revenue are in model units. Shifting costs are fixed within each reputation comparison. The discount factor is 0.6. Every row satisfies both announcement constraints and the implementation condition.

5.4 Numerical Illustration

I solve the two-period model with costs deductible in both countries, retaining the production parameters of Figure 2 and using $\beta = 0.6$. At each prior, I first solve the selected one-period problem for the continuation. The initial tax then maximizes revenue subject to the implementation constraint and both announcement constraints.

For $\gamma = 1.25$, moving p from 0.1 to 0.9 reduces the sustainable rate from 45.4 to 35.8 percent. Capital increases from 2.510 to 3.272 (Table 5). At the two priors, the discounted loss of future revenue from reneging is 0.0008 and 0.0809, respectively. Because p does not affect the current revision gain at a fixed tax, the larger loss is what allows the government to reduce its rate. Each of the other cost comparisons also has a lower implemented rate and higher first-period investment at the better reputation.

Figure 4 compares discount factors $\beta = 0.3, 0.6, 0.9$ at fixed shifting costs. Greater patience lowers the sustainable rate and increases capital at each reputation shown. For example, at $p = 0.5$, moving β from 0.3 to 0.9 lowers the rate from 42.5 to 39.7 percent, with investment increasing from 2.727 to 2.954. Once full commitment is attainable, further patience no longer reduces the tax.

5.5 Profit-Shifting Costs and the Incentive to Renege

A reduction in shifting costs gives firms more opportunity to protect their returns under an opportunistic government. For the current cohort, this reduces the revenue from a tax increase on installed capital. For future cohorts, it can support more investment even after the government has been revealed as opportunistic. The latter response raises revenue following a loss of reputation and makes reneging less costly. I recompute both revenue differences to determine which effect governs the sustainable tax.

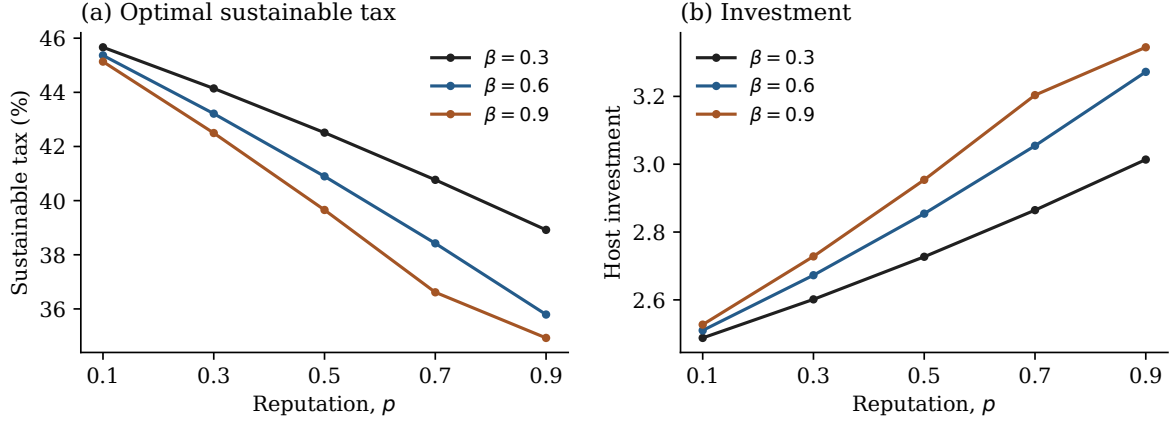


FIGURE 4: Government patience, sustainable taxation, and investment

Notes: Both types honor the first-period tax in every displayed allocation. The discount factor varies across curves. Other parameters equal those in Figure 2, with $\gamma = 1.25$. The continuation in each case is the selected one-period equilibrium. The full-commitment tax does not depend on the discount factor or the prior.

TABLE 6: Higher shifting costs and sustainable taxation

p	Revenue change		Optimal tax (%)	
	Current gain	Future loss	$\gamma = 1.25$	$\gamma = 1.5$
0.1	0.006090	0.000045	34.29	36.68
0.5	0.012397	0.001303	32.43	34.48
0.9	0.018407	0.004367	30.49	32.18
Full commitment	—	—	30.32	31.99

Notes: Current gain and future loss are changes in the two sides of (13) when γ increases from 1.25 to 1.5, evaluated at the initial tax optimum for $\gamma = 1.25$. Future loss includes the discount factor. Revenue changes are in model units, and tax rates are in percent. The last row removes the implementation constraint and solves each full-commitment problem. Costs are deductible in the country of origin. Both cost values satisfy (9). Other parameters are $z = 1$, $\alpha = 0.66$, $r^* = 0.04$, $\bar{k} = 10$, and $\tau^* = 0.15$.

I use origin-country costs for this exercise, with $\beta = 0.6$ and the same production parameters, and raise γ from 1.25 to 1.5. First, I fix the initial tax at the optimum for $\gamma = 1.25$. At the new cost, I solve again for optimal choices of investment, profit shifting, tax revision, and the two continuation values. This gives the changes in the current gain and future loss at the old tax. I then reoptimize the initial tax to obtain the last column of Table 6.

Both revenue differences rise in Table 6. The increase in the current gain is larger than the increase in the discounted future loss, so the government would renege on the original tax. It chooses a higher sustainable rate instead. At $p = 0.5$, that rate rises from 32.4 to 34.5 percent. The same ordering holds for the other reported priors. Reversing this cost change would allow more shifting and result in a lower sustainable tax.

Under full commitment, the tax increase for this change in costs is 1.66 percentage points. With limited commitment, it is 2.39 points at reputation 0.1, 2.05 points at 0.5, and 1.69 points at 0.9. Thus, an initially poor reputation requires a larger increase beyond the full-commitment response. Little continuation revenue is lost by revealing the opportunistic type in that case, making it harder to deter a tax revision when it becomes more profitable. Appendix B.5 gives the calculation.

The comparison also holds when I vary patience over $\beta = 0.3, 0.6, 0.9$ and include a cost increase from $\gamma = 0.5$ to 1.25. In all eighteen cases, the current gain rises more than the discounted future loss. The tax increase under limited commitment is at least as large as under full commitment, exceeding it by up to 2.33 percentage points. With $p = 0.9$ and $\beta = 0.9$, full commitment is attainable at either cost, and the increases coincide. Appendix B.5 reports the complete set of comparisons.

Reporting restrictions therefore affect corporate taxation through the incentive to maintain policy as well as the investment and tax-base responses emphasized by Hong and Smart (2010) and Mongrain et al. (2023). Comparing the same cost change under full and limited commitment shows why its tax consequences can differ across governments with different reputations.

6 Concluding Remarks

Profit shifting changes the tax response to government reputation. It protects firms' returns after capital is installed and sustains investment under uncertain commitment. With sufficient protection, better reputation makes a tax cut more effective at attracting capital. The government lowers its tax under commitment while investment and revenue rise. Without profit shifting, the optimal tax instead rises. Numerical comparisons extend the declining response to other shifting costs and foreign taxes.

The relevant protection is the transfer firms could make after a large tax increase. A restriction can make confiscation optimal without changing observed transfers or their response to small tax differences. Future investment provides additional discipline. Better reputation increases the revenue lost after reneging and permits a lower sustainable tax, including in the finite-horizon extension. In the numerical cost comparisons, higher shifting costs increase the current gain by more than the future loss. The sustainable tax then rises more than the full-commitment tax, especially at low reputation.

The data associate higher quality of government with higher statutory taxes across countries, greater inward FDI within countries, and more net inward shifting conditional on taxation. The model's tax response is positive with limited protection against a revision and negative with sufficient shifting capacity. Distinguishing these cases

empirically requires information about available reporting adjustments beyond the taxes and transfers observed under current policy.

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A Analytical Results and Proofs

A.1 Profit Shifting and Investment

The notation follows Section 3. Aggregate profits and transfers are $\Lambda(K) = \rho(K)$ and $\Theta(K, \tau) = \theta(K, \tau)$ because firms have unit measure. Parameters not displayed as function arguments are held fixed.

Solving the two nonnegative-base inequalities gives

$$\begin{aligned} \theta &\geq \max \left\{ \frac{\rho(k)(1 - \sqrt{1 + 2\gamma})}{\gamma}, \frac{\rho(\bar{k} - k)(-1 - \sqrt{1 + 2\gamma})}{\gamma} \right\}, \\ \theta &\leq \min \left\{ \frac{\rho(k)(1 + \sqrt{1 + 2\gamma})}{\gamma}, \frac{\rho(\bar{k} - k)(-1 + \sqrt{1 + 2\gamma})}{\gamma} \right\}. \end{aligned} \quad (15)$$

The firm's payoff is strictly concave in the transfer. Its optimum is therefore the point in this interval closest to the unconstrained transfer in (5). At zero profits in either country, only a zero transfer is feasible. On the interior, substituting the optimal transfer gives host revenue

$$\tau \rho(K) \left\{ 1 + \frac{1}{2\gamma} - \frac{1}{2\gamma} \left[\frac{(1 - \tau^*)[\rho(K) + \rho(\bar{k} - K)]}{(1 - \tau)\rho(\bar{k} - K) + (1 - \tau^*)\rho(K)} \right]^2 \right\}. \quad (16)$$

Its second derivative with respect to the tax is

$$\begin{aligned} & - \frac{\rho(K)(1 - \tau^*)^2[\rho(K) + \rho(\bar{k} - K)]^2\rho(\bar{k} - K)}{\gamma} \\ & \times \left\{ \frac{2}{[(1 - \tau)\rho(\bar{k} - K) + (1 - \tau^*)\rho(K)]^3} + \frac{3\tau\rho(\bar{k} - K)}{[(1 - \tau)\rho(\bar{k} - K) + (1 - \tau^*)\rho(K)]^4} \right\} < 0. \end{aligned} \quad (17)$$

Thus revenue is strictly concave over the interval with nonbinding reporting constraints. A binding foreign-base constraint gives a fixed positive host base and revenue increasing linearly with the tax. At the transition to the interior, the revenue derivative falls. A binding host-base constraint gives zero revenue, which cannot dominate a positive interior base. The government consequently has a unique tax best response at every positive interior capital allocation, including a possible maximum at one.

For investment, the function $-\theta^2/\rho(k)$ is jointly concave in positive profit and the transfer and increases with profit. Composition with the concave production profit preserves concavity. Both taxable bases are therefore concave in capital and the transfer, and requiring nonnegative bases gives a convex feasible set. The weighted sum in the firm's objective is strictly concave between distinct capital choices because

$1 - \tau^* > 0$ and foreign profits are strictly concave. Maximizing over transfers preserves strict concavity in capital. The individual investment optimum is unique for given implemented taxes. This does not establish uniqueness of the aggregate investment and tax fixed point.

At interior investment with nonbinding profit-shifting constraints, the first-order condition equates expected after-tax marginal profits,

$$\begin{aligned} \mathbb{E}_\tau \left[(1 - \tau) \left(1 + \frac{\gamma \theta(k, \tau)^2}{2\rho(k)^2} \right) \rho'(k) \right] \\ = (1 - \tau^*) \mathbb{E}_\tau \left[\left(1 + \frac{\gamma \theta(k, \tau)^2}{2\rho(\bar{k} - k)^2} \right) \rho'(\bar{k} - k) \right]. \end{aligned} \quad (18)$$

A.2 Equilibrium and Announcement Incentives

An equilibrium with a known opportunistic government consists of mutual investment and tax best responses. With positive shifting costs, their payoffs extend continuously to the compact capital and tax intervals under the constrained reporting rule. The individual investment best response is unique. The tax best response is unique at interior capital, all taxes are optimal at zero host capital, and a tax of one is optimal when all capital is in the host. Both correspondences have nonempty compact convex values and are upper hemicontinuous. Kakutani's theorem gives a fixed point, without establishing its uniqueness.

For the pooling equilibrium, firms retain prior p after the prescribed announcement. Each firm maximizes (7) while taking the implemented taxes as given. The commitment type implements its announcement and the opportunistic type solves (6). After an unexpected announcement, beliefs assign probability one to the opportunistic type and investment is the capital in the chosen known-type equilibrium. The largest revenue either type could obtain after such an announcement is the revenue-maximizing tax on that capital. The two explicit inequalities in (8) require each type's equilibrium revenue to be at least this amount and prevent an announcement deviation. Firms optimize profit shifting after all subsequent histories. Bayes' rule does not restrict beliefs after a zero-probability announcement. These strategies and beliefs define a weak perfect Bayesian equilibrium. The off-path belief can be approached by fully supported announcement trembles in which the commitment type's deviation probability vanishes faster than the opportunistic type's. The on-path posterior converges to the prior. The revenue selection in Section 3.5 ranks its supported common announcements.

A.3 Net Profits and the Investment Response

Let $\rho(k) = zk^\alpha - r^*k$ and maintain $\alpha z\bar{k}^{\alpha-1} > (2 - \alpha)r^*$. Marginal profits are positive on $(0, \bar{k}]$. Differentiating the investment condition shows that the elasticity of host profits with respect to the ratio of retained returns is

$$\frac{\rho'(K)}{\rho(K) \left[-\frac{\rho''(K)}{\rho'(K)} - \frac{\rho''(\bar{k} - K)}{\rho'(\bar{k} - K)} \right]}. \quad (19)$$

This expression decreases strictly with K . To see this, write its reciprocal as the sum of $-\rho(K)\rho''(K)/\rho'(K)^2$ and $[\rho(K)/\rho'(K)][-\rho''(\bar{k} - K)/\rho'(\bar{k} - K)]$. The first term increases when $r^* > 0$ and is constant when $r^* = 0$. In the second term, the first factor increases by concavity. The second factor increases with host capital because

$$-\frac{\frac{d}{dk} \left[-\frac{\rho''(k)}{\rho'(k)} \right]}{\left[-\frac{\rho''(k)}{\rho'(k)} \right]^2} = \frac{1 - (2 - \alpha)r^*/(\alpha z k^{\alpha-1})}{1 - \alpha} > 0. \quad (20)$$

The reciprocal of (19) therefore increases strictly. It tends to $(1 - \alpha)/\alpha$ as K tends to zero.

For $K \leq \bar{k}/2$, the rate at which this responsiveness decreases satisfies

$$\begin{aligned} & -\frac{1}{-\rho''(K)/\rho'(K) - \rho''(\bar{k} - K)/\rho'(\bar{k} - K)} \\ & \times \frac{d}{dK} \log \left\{ \frac{\rho'(K)}{\rho(K) \left[-\rho''(K)/\rho'(K) - \rho''(\bar{k} - K)/\rho'(\bar{k} - K) \right]} \right\} \\ & \leq \frac{1}{1 - \alpha} \left[\frac{1}{2} + \frac{(2 - \alpha)r^*}{\alpha z (\bar{k}/2)^{\alpha-1}} \right]. \end{aligned} \quad (21)$$

To obtain the bound, use (20) and

$$\frac{\rho'(K)/\rho(K)}{-\rho''(K)/\rho'(K)} = \frac{\alpha}{1 - \alpha} \frac{[1 - r^*/(\alpha z K^{\alpha-1})]^2}{1 - r^*/(z K^{\alpha-1})}.$$

Substitution into the logarithmic derivative bounds its left side in (21) above by

$$\begin{aligned} & \frac{1}{1 - \alpha} \left\{ \frac{-\rho''(\bar{k} - K)/\rho'(\bar{k} - K)}{-\rho''(K)/\rho'(K) - \rho''(\bar{k} - K)/\rho'(\bar{k} - K)} \right. \\ & \left. + (2 - \alpha) \left[\frac{-\rho''(K)/\rho'(K)}{-\rho''(K)/\rho'(K) - \rho''(\bar{k} - K)/\rho'(\bar{k} - K)} \right]^2 \frac{r^*}{\alpha z K^{\alpha-1}} \right\}. \end{aligned}$$

Equation (20) implies that host curvature is at least foreign curvature when $K \leq \bar{k}/2$. The first fraction is therefore at most one half, and the squared fraction is at most one. Finally, $K^{1-\alpha} \leq (\bar{k}/2)^{1-\alpha}$ gives (21). The same formulas imply that (19) is at most $\alpha/(1-\alpha)$.

A.4 Origin-Country Costs and Retained Returns

With origin-country costs and $\rho(k) = zk^\alpha - r^*k$, an outward transfer has the interior solution

$$\frac{-\Theta(K, \tau)}{\Lambda(K)} = \frac{\tau - \tau^*}{\gamma(1 - \tau)}, \quad 0 \leq \frac{-\Theta(K, \tau)}{\Lambda(K)} \leq \frac{\sqrt{1 + 2\gamma} - 1}{\gamma}. \quad (22)$$

The upper bound follows from the nonnegative host base. Before it binds, firms retain total after-tax profits

$$\left[(1 - \tau) + \frac{(\tau - \tau^*)^2}{2\gamma(1 - \tau)} \right] \rho(K) + (1 - \tau^*)\rho(\bar{k} - K). \quad (23)$$

The interior revenue-maximizing tax τ^O is independent of capital and solves

$$(2\gamma + 1)(1 - \tau^O)^3 = (1 - \tau^*)^2(1 + \tau^O). \quad (24)$$

The investment $K(\tau^O, 0)$ denotes equilibrium investment when firms know that this tax will be implemented. It is obtained from the marginal-profit condition

$$\left[(1 - \tau^O) + \frac{(\tau^O - \tau^*)^2}{2\gamma(1 - \tau^O)} \right] \rho'(K) = (1 - \tau^*)\rho'(\bar{k} - K). \quad (25)$$

Before the host-base constraint binds, the ratio of retained host to foreign returns satisfies

$$\frac{(1 - \tau) + (\tau - \tau^*)^2/[2\gamma(1 - \tau)]}{1 - \tau^*} \geq \frac{2}{\sqrt{1 + 2\gamma} + 1}. \quad (26)$$

To obtain this bound, differentiate the numerator with respect to the tax. The derivative is minus the host base per unit of profit, so the minimum occurs when this base reaches zero. Substituting the transfer at that point gives the right side. Once the host-base constraint binds, the transfer share remains $2/(\sqrt{1 + 2\gamma} + 1)$, the retained host return equals $(1 - \tau^*)2/(\sqrt{1 + 2\gamma} + 1)$, and host revenue is zero. The same lower bound therefore holds up to a tax of one, using this constrained payoff rather than the interior formula.

The fourth restriction in (9) ensures that the retained return under discretion is large enough relative to the responsiveness bound in (21). The second and third restrictions exclude optimal taxes below τ^* as shown next.

A.5 Proof of Proposition 1

For an outward transfer, revenue per unit of host profits is

$$\tau \left[1 - \frac{\tau - \tau^*}{\gamma(1 - \tau)} - \frac{(\tau - \tau^*)^2}{2\gamma(1 - \tau)^2} \right]. \quad (27)$$

Its second derivative is $-(1 - \tau^*)^2(\tau + 2)/[\gamma(1 - \tau)^4] < 0$. Its first derivative at τ^* is $1 - \tau^*/[\gamma(1 - \tau^*)] > 0$. Revenue becomes zero when the host-base bound is reached. There is consequently a unique maximum above τ^* and below that bound, given by (24). The coefficient on host profits in the firm's retained return decreases over the positive-base interval. Its derivative is minus the host taxable base per unit of profit.

For $\tau < \tau^*$, inward shifting deducts its cost abroad and the transfer equals $(\tau^* - \tau)\rho(\bar{k} - K)/[\gamma(1 - \tau^*)]$. The second restriction in (9) makes this transfer feasible throughout the inward interval. Firm profits then equal

$$(1 - \tau)\rho(K) + \left[(1 - \tau^*) + \frac{(\tau^* - \tau)^2}{2\gamma(1 - \tau^*)} \right] \rho(\bar{k} - K).$$

The host return coefficient is at least the foreign coefficient. Under outward shifting, their ratio is at least $(\sqrt{1 + 2\gamma} - 1)/\gamma$. These bounds also hold under uncertainty about which tax is implemented. When $K \leq \bar{k} - K$, subtracting the rental cost gives $\rho'(K)/\rho'(\bar{k} - K) \geq [K/(\bar{k} - K)]^{\alpha-1}$ and $\rho(K)/\rho(\bar{k} - K) \geq [K/(\bar{k} - K)]^\alpha$. The investment condition and the retained-return bound therefore imply

$$\frac{\rho(K)}{\rho(\bar{k} - K)} \geq \left(\frac{\sqrt{1 + 2\gamma} - 1}{\gamma} \right)^{\alpha/(1-\alpha)}.$$

If $K > \bar{k} - K$, the same lower bound holds because the profit ratio exceeds one. For fixed such capital, the derivative of revenue on the inward interval is $\rho(K) + (\tau^* - 2\tau)\rho(\bar{k} - K)/[\gamma(1 - \tau^*)] > 0$ by the third restriction. No inward tax improves on τ^* . The global tax best response is therefore the same τ^O at every equilibrium capital allocation.

For announcements between τ^* and τ^O , investment satisfies

$$\begin{aligned} & \left\{ p \left[(1 - \tau^R) + \frac{(\tau^R - \tau^*)^2}{2\gamma(1 - \tau^R)} \right] + (1 - p) \left[(1 - \tau^O) + \frac{(\tau^O - \tau^*)^2}{2\gamma(1 - \tau^O)} \right] \right\} \rho'(K) \\ & = (1 - \tau^*)\rho'(\bar{k} - K). \end{aligned} \quad (28)$$

Capital increases with p and decreases with the announcement. An announcement above τ^O reduces both the firm's retained return and the government's revenue per unit of profit, so it is dominated by τ^O .

To exclude announcements below τ^* , differentiate the investment condition using the inward-transfer payoff. Its host coefficient is $p(1 - \tau^R) + (1 - p)[(1 - \tau^O) + (\tau^O - \tau^*)^2 / (2\gamma(1 - \tau^O))]$ and its foreign coefficient is $(1 - \tau^*) + p(\tau^* - \tau^R)^2 / [2\gamma(1 - \tau^*)]$. The second restriction implies that their ratio decreases with τ^R . Thus investment exceeds $K(\tau^O, 0)$ throughout this interval. Log differentiation also gives

$$\frac{\partial \log \rho(K)}{\partial \tau^R} \geq -\frac{\alpha}{(1 - \alpha)(1 - \tau^*)}.$$

Using the bound on the profit ratio, the derivative of honored revenue divided by host profits is at least

$$1 - \frac{\tau^*}{\gamma(1 - \tau^*)} \left(\frac{\sqrt{1 + 2\gamma} + 1}{2} \right)^{\alpha/(1 - \alpha)} - \frac{\tau^* \alpha}{(1 - \alpha)(1 - \tau^*)} > 0,$$

where the strict inequality is the third restriction in (9). Revenue therefore increases below τ^* and immediately to its right. At τ^O , its derivative is negative when $p > 0$, since the direct revenue derivative is zero and capital falls with the announcement. All maximizing announcements lie strictly between the two rates.

It remains to order these interior maxima. Log differentiation of honored revenue gives the derivative of the log of (27), which is independent of p , plus the derivative of the retained-return coefficient, which is negative, multiplied by

$$\frac{p\rho'(K)^2}{(1 - \tau^*)\rho'(\bar{k} - K)\rho(K) \left[-\frac{\rho''(K)}{\rho'(K)} - \frac{\rho''(\bar{k} - K)}{\rho'(\bar{k} - K)} \right]}. \quad (29)$$

The bound in (21) implies that the derivative of this factor with respect to p , after division by positive terms, is bounded below by

$$(1 - \tau^O) + \frac{(\tau^O - \tau^*)^2}{2\gamma(1 - \tau^O)} - \frac{1}{1 - \alpha} \left[\frac{1}{2} + \frac{(2 - \alpha)r^*}{\alpha z(\bar{k}/2)^{\alpha-1}} \right] \left[(1 - \tau^*) - (1 - \tau^O) - \frac{(\tau^O - \tau^*)^2}{2\gamma(1 - \tau^O)} \right] > 0. \quad (30)$$

To obtain this bound, differentiate the product of p , the investment responsiveness in (19), and the reciprocal of the expected retained-return coefficient. The direct effect of p contributes the retained return under discretion. The capital response contributes at worst minus p times the bound in (21) times the difference between the commitment and discretionary retained returns. That difference is at most the foreign retained return minus the discretionary retained return. The fourth restriction in (9) and (26) make the displayed lower bound positive. Log revenue consequently has strictly decreas-

ing differences in the announcement and reputation. Adding the two maximization inequalities excludes a higher announcement at higher reputation. Equality is excluded by the interior first-order conditions, since the log-revenue derivative falls strictly with reputation. This orders every maximizing announcement without assuming uniqueness.

A higher p and lower announcement both raise investment. Honored revenue rises strictly because it rises with p at each fixed announcement below τ^O . Dividing this higher revenue by the lower tax gives a larger honored base. Opportunistic revenue also rises because (27) at τ^O is a fixed positive coefficient multiplying host profits. Announcing τ^O reproduces the known-type allocation at any prior. The selected announcement yields at least that honored revenue and more investment, so both types' announcement constraints hold. The stated off-path beliefs support these allocations for $0 < p < 1$. At $p = 1$, the selected announcement is the full-commitment optimum.

Finally, the tax and outward share under commitment are below their opportunistic counterparts and decrease with reputation. The expected implemented rate and expected outward share,

$$p\tau^R + (1 - p)\tau^O, \quad -\frac{p\Theta(K, \tau^R) + (1 - p)\Theta(K, \tau^O)}{\Lambda(K)}, \quad (31)$$

therefore decrease strictly. The outward share in (22) does not depend on capital. The amount shifted additionally depends on the expanding host profits.

A.6 Equilibrium Selection

This subsection studies pure announcement strategies under the assumptions of Proposition 1. The discretionary tax is the same τ^O at every investment induced by an announcement and a belief. The allocation after an unexpected announcement is therefore uniquely determined when the government is believed to be opportunistic. These properties allow the announcement incentives to be compared across equilibria.

I apply the strongly coherent equilibrium criterion of Lu (2013). It evaluates an unexpected announcement using the payoffs of an alternative equilibrium in which that announcement is made. A nonempty subset of its senders must weakly prefer the alternative, with at least one strict preference. Within that subset, a strictly improving type sends with probability one, an indifferent type with probability between zero and one, and an excluded type with probability zero. Bayes' rule then gives the associated belief. A candidate fails the criterion if an alternative announcement admits such beliefs about a type, but none can support the candidate. This comparison concerns

incentives across equilibria, rather than gains from a deviation under an arbitrary receiver response.

Lemma 1 (Selection of the tax under commitment). *Under the assumptions of Proposition 1, fix $p \in (0, 1)$ and restrict announcement strategies to pure strategies. The strongly coherent equilibrium outcomes are pooling outcomes that maximize the commitment type's revenue, as in (8). Every such maximizing outcome can be supported by belief zero after any unexpected announcement.*

Proof. Announcing τ^O gives both government types revenue $R(0, \tau^O)$ at every posterior belief. Firms then face τ^O with certainty. Consequently, the commitment type can secure that revenue regardless of the beliefs assigned to an unexpected announcement.

Consider an equilibrium with different announcements by the two types. The opportunistic type is revealed by its announcement and receives $R(0, \tau^O)$. If the commitment type announces a tax below τ^O , firms invest strictly more than $K(\tau^O, 0)$. This follows from the declining ratio of retained host to foreign returns on $[0, \tau^O]$, established in Appendix A.5. Mimicking that announcement and imposing τ^O would therefore give the opportunistic type strictly more revenue. If the commitment type announces a tax above τ^O , its revenue is strictly below $R(0, \tau^O)$. Both investment and revenue per unit of host profits are lower, or its tax base is zero. The commitment type must thus announce τ^O in any separating equilibrium, and both types receive exactly $R(0, \tau^O)$.

Proposition 1 gives both types revenue strictly above $R(0, \tau^O)$ at a maximizing pooling announcement. The commitment type has an interior maximum and a negative revenue derivative at τ^O . The opportunistic type gains from higher investment. By continuity, these inequalities also hold at nearby announcements. Choose one outside the path of a candidate separating equilibrium. The associated posterior is at least p under any strongly coherent belief about the commitment type. Specifically, it is p when both strictly improving types send, and one when only the commitment type sends. The commitment type's revenue is increasing in this posterior, so the gain at the prior remains a gain under either belief. No separating candidate can therefore satisfy the criterion.

Next, suppose the candidate is pooling but fails to maximize the commitment type's revenue. A maximizing announcement belongs to another equilibrium and is off path in the candidate. The commitment type strictly prefers this alternative, so any associated strongly coherent posterior is at least p . If the opportunistic type also gains strictly, including it gives p . A loss excludes it, while indifference allows a sending probability between zero and one. All such beliefs leave the commitment type at least as well

off as at the prior, where the improvement is strict. Hence, the criterion excludes this candidate too.

It remains to support a maximizing pooling allocation. Assign probability zero to the commitment after any unexpected announcement. Proposition 1 supplies both announcement constraints. Among alternative pure equilibria, none strictly improves the commitment type's payoff, and separating alternatives reduce both types' payoffs. If a pooling alternative strictly improves a type's payoff, that type can only be opportunistic. The proposed belief is then strongly coherent by assigning sending probability zero to the commitment type, including when it is indifferent. If neither type strictly improves, the criterion does not restrict beliefs. Hence, these off-path beliefs satisfy the criterion and support every maximizing pooling allocation. $\square$

The criterion thus selects the maximizing pooling outcomes of the commitment type. No further selection is needed for the comparative static in Proposition 1, since it orders every maximizing tax at the two priors.

This selection differs from the intuitive criterion of [Cho and Kreps \(1987\)](#), which the selected pooling allocation fails to meet. Consider increasing the announcement toward τ^O until investment at belief one equals investment in the original allocation. The required rate is strictly between $\tau^R(p)$ and τ^O . The opportunistic type would then receive the same revenue as before, while the commitment type would earn more because revenue per unit of profit rises below τ^O . A slightly higher announcement makes the opportunistic type worse off even at belief one, where investment is greatest, but still benefits the commitment type at that belief. The intuitive criterion assigns this deviation to the commitment type, validating its gain. The criterion in [Lu \(2013\)](#) instead uses alternative equilibria to evaluate an announcement. It is this restriction on the comparison specified in the lemma that yields the selection above.

A.7 Proof of Proposition 2

Without profit shifting, (10) implies $K_{\tau^R} < 0$ and $K_p > 0$. Differentiating the government's objective $\tau^R \rho(K)$ gives

$$\tau^R(p) = 1 - \left\{ 1 + \left[-\frac{\rho''(K)}{\rho'(K)} - \frac{\rho''(\bar{k} - K)}{\rho'(\bar{k} - K)} \right] \frac{\rho(K)}{\rho'(K)} \right\}^{-1}. \quad (32)$$

By Appendix A.3, the expression in braces increases strictly with K . Substituting this tax into

$$p = \frac{1 - \tau^* \frac{\rho'(\bar{k} - K)}{\rho'(K)}}{1 - \tau^R}$$

gives a strictly increasing function of capital. It tends to zero as K tends to zero and to infinity as K tends to $\bar{k}$. There is exactly one stationary tax for every $p > 0$, and both capital and this tax increase with p . Revenue is positive at interior taxes and zero at zero and one, so the unique stationary tax is the global maximizer. A known opportunistic government attracts no capital and earns no revenue. Each type's positive revenue at the selected announcement therefore satisfies its announcement constraint. Higher capital in (10) also requires a higher retained-return coefficient $p(1 - \tau^R)$. The expected implemented tax $1 - p(1 - \tau^R)$ therefore decreases.

More generally, this argument applies to any three-times continuously differentiable profit function with positive marginal profits, strict concavity, an infinite marginal profit at zero, and an expression in braces in (32) that is strictly increasing and bounded near zero. The rental-cost specification satisfies these requirements under the condition stated in Proposition 2.

A.8 Proof of Proposition 3

Maintain the assumptions and transfer-share domain in Proposition 3. The investment condition established in the proof of Proposition 1 implies

$$\frac{\rho(K)}{\rho(\bar{k} - K)} \geq \left(\frac{2}{\sqrt{1 + 2\gamma} + 1} \right)^{\alpha/(1-\alpha)} > \frac{\tau^*}{\gamma(1 - \tau^*)}. \quad (33)$$

The strict inequality follows from the third primitive restriction in (9). On the inward-shifting interval, the revenue derivative is

$$\rho(K) + \frac{(\tau^* - 2\tau)\rho(\bar{k} - K)}{\gamma(1 - \tau^*)} \geq 0,$$

with strict inequality for $\tau < \tau^*$. Thus no inward-shifting tax dominates τ^* . On the outward interval without the additional limit, (27) is strictly concave and has its unique maximum at $\tau^O > \tau^*$.

Below the tax at which the transfer limit binds, revenue equals host profits times (27). Above that tax, the base no longer changes, so revenue is linear in the rate and increases whenever the base is positive. There are two cases to consider. If the limit binds before the unrestricted revenue maximum, revenue is increasing on both sides of the point at which it binds, and a tax of one maximizes revenue. If it binds later, the government compares the unrestricted interior maximum with the revenue at a tax of one. Equating the two gives (11).

At the unrestricted transfer, the base at a tax of one exceeds the revenue collected at $\tau^O < 1$. The critical share is therefore strictly larger than that unrestricted transfer share.

It is strictly smaller than the nonnegative-base bound because the unrestricted optimum collects positive revenue. At the threshold the two candidates tie. For every positive limit, a neighborhood of equal tax rates remains unconstrained, and the derivative of the outward share with respect to the host tax is $1/[\gamma(1 - \tau^*)]$. The limit changes the global tax comparison while preserving that derivative.

A.9 Proof of Proposition 4

Maintain the origin-country cost specification and (9). Proposition 1 gives a discretionary tax τ^O that does not depend on capital. It also establishes unique investment $K(\tau^O, 0)$ when the type is known and continuation revenue that increases strictly with reputation. I first establish the properties of current revenue and the gain from reneging.

Under certain implementation, investment satisfies (28) at $p = 1$. For $\tau^* < \tau < \tau^O$, differentiation gives

$$\begin{aligned} \frac{\partial \log \rho(K(\tau, 1))}{\partial \tau} = & - \frac{\rho'(K)}{\rho(K) \left[-\frac{\rho''(K)}{\rho'(K)} - \frac{\rho''(\bar{k} - K)}{\rho'(\bar{k} - K)} \right]} \\ & \times \frac{1 - \frac{\tau - \tau^*}{\gamma(1 - \tau)} - \frac{(\tau - \tau^*)^2}{2\gamma(1 - \tau)^2}}{(1 - \tau) + \frac{(\tau - \tau^*)^2}{2\gamma(1 - \tau)}} < 0. \end{aligned} \quad (34)$$

Here $K = K(\tau, 1)$. Current revenue is (27) times $\rho(K(\tau, 1))$. Its derivative is positive at τ^* and negative at τ^O , as established in the proof of Proposition 1. At any stationary tax between these rates, log differentiation and substitution of the first-order condition yield

$$\frac{\partial^2 \log R(1, \tau)}{\partial \tau^2} < - \frac{(1 - \tau^*)^2(1 + 2\tau)}{\gamma\tau(1 - \tau)^4 \left[1 - \frac{\tau - \tau^*}{\gamma(1 - \tau)} - \frac{(\tau - \tau^*)^2}{2\gamma(1 - \tau)^2} \right]} < 0. \quad (35)$$

To obtain the bound, use the stationarity condition to substitute the percentage increase in revenue per unit of host profit for the percentage decline in host profits. The former equals $1/\tau$ minus the second derivative of the retained-return coefficient divided by the host base per unit of profit. The responsiveness in (19) decreases with capital by Appendix A.3. Its derivative and the remaining squared terms in the second derivative of log revenue are negative. Thus every stationary tax in this interval is a strict maximum. The endpoint signs imply that there is exactly one, with revenue increasing before it and decreasing afterward. Below τ^* revenue increases, and above τ^O both the retained host return and revenue per unit of profit fall until the host base

reaches zero. Revenue is zero thereafter. It follows that $\tau^R(1) \in (\tau^*, \tau^O)$ is the unique global maximizer, and revenue is strictly decreasing from $\tau^R(1)$ to τ^O .

The gain from reneging is continuous. For $\tau_1^R \in [\tau^*, \tau^O]$, it is host profits $\rho(K(\tau_1^R, 1))$ times the difference between revenue per unit of profit in (27) evaluated at τ^O and at τ_1^R . Both positive factors decrease strictly with τ_1^R , so the gain decreases to zero at τ^O . Below τ^* , certain-tax investment decreases by the second restriction in (9), while honored revenue increases by the third restriction. Revenue after the best revision is the fixed positive coefficient (27) evaluated at τ^O times these decreasing host profits. The difference between revision revenue and honored revenue therefore decreases there as well. The gain is consequently strictly decreasing on the entire interval $[0, \tau^O]$. Taxes above τ^O yield less current revenue than $R(0, \tau^O)$, since both capital and revenue per unit of profit are lower, or the host base is zero. These conclusions derive the required revenue properties from (9).

Now fix $p \in (0, 1)$ and the selected continuation equilibrium. Both types' continuation revenues are at least $R(0, \tau^O)$ by their one-period announcement constraints. The opportunistic continuation revenue is strictly greater because investment exceeds the known-type allocation. The right side of (13) is therefore positive. Any first-period tax earning at least $R(0, \tau^O)$ also satisfies both announcement constraints in (14).

At the certain tax τ^O , first-period capital and revenue equal their known-type values, and there is no gain from a revision. The full-commitment tax earns strictly greater current revenue. If it satisfies (13), it also satisfies the announcement constraints and is the unique optimum. Otherwise, the revision gain at $\tau^R(1)$ exceeds the continuation loss. Since that gain is continuous and strictly decreasing to zero at τ^O , a unique rate in $(\tau^R(1), \tau^O)$ satisfies the constraint with equality. Every lower rate violates implementation. Every higher rate up to τ^O yields lower current revenue, and a rate above τ^O yields less than $R(0, \tau^O)$. The binding rate is therefore the unique optimum. Its current revenue exceeds $R(0, \tau^O)$, which verifies the two announcement constraints.

An increase in p raises the continuation loss by Proposition 1. In (13), investment under certain implementation and the gain from a current revision remain fixed for a given τ_1^R , regardless of p . A binding constraint is therefore relaxed. The revenue-maximizing tax moves down toward $\tau^R(1)$, declining strictly whenever its original value is above $\tau^R(1)$. If it already equals $\tau^R(1)$, the unconstrained optimum remains feasible and is unchanged. By (34), every strict rate reduction increases investment.

For a rise in β , holding p and the continuation selection fixed, the right side of (13) increases strictly. Applying the same argument gives a weak reduction in the optimal tax, strict when its starting value exceeds $\tau^R(1)$. The resulting current and continuation revenues each remain no smaller than $R(0, \tau^O)$, so the two announcement constraints are preserved.

A.10 Taxation over a Finite Horizon

Consider $T \geq 2$ periods in the environment of Section 5, with a new cohort investing each period and no state connecting cohorts other than beliefs. Government type stays fixed, and both types discount revenues by $\beta \in (0, 1)$. I consider pooling play with implementation by both types before period T and the selected one-period equilibrium at the terminal date. Following either an unexpected announcement or a tax revision, firms assign belief zero and play the known-opportunistic allocation at every subsequent date. After a history with no deviation, the selected policy maximizes the commitment type's revenue among these equilibria.

In the origin-country specification, the discretionary tax is τ^O in every period. The implementation constraint at date $t < T$ is

$$\begin{aligned} & R(1, \tau_t^O) - R(1, \tau_t^R) \\ & \leq \sum_{s=t+1}^{T-1} \beta^{s-t} \left[R(1, \tau_s^R) - R(0, \tau^O) \right] + \beta^{T-t} \left[R(p, \tau_T^{RO}) - R(0, \tau^O) \right]. \end{aligned} \quad (36)$$

The sum is zero when $t = T - 1$. Each current revenue uses the capital induced by that period's announcement. The terminal revenue instead uses $K(\tau^R(p), p)$. The right side is the discounted revenue lost when subsequent cohorts learn the government's type.

Proposition A.1 (Reputation and taxes before the final period). *Suppose the assumptions of Proposition 4 hold and consider the T -period equilibria just described. For every $t < T$, the selected tax is unique and belongs to $[\tau^R(1), \tau^O]$. For $0 < p < p' < 1$,*

$$\tau_t^R(p') \leq \tau_t^R(p), \quad \text{strictly if } \tau_t^R(p) > \tau^R(1).$$

Every selected tax satisfies the implementation constraint and both types' announcement incentives.

Proof. Proposition 1 establishes that terminal opportunistic revenue exceeds $R(0, \tau^O)$ and rises strictly with p . Terminal commitment revenue also exceeds $R(0, \tau^O)$. Appendix A.9 establishes that current revenue has its unique maximum at $\tau^R(1)$ and decreases up to τ^O , while the gain from reneging decreases strictly to zero on $[0, \tau^O]$.

At $t = T - 1$, the conclusion is Proposition 4. The selected current revenue exceeds $R(0, \tau^O)$ and weakly increases with p , since a higher reputation lowers the selected tax toward its revenue maximum. Adding discounted terminal opportunistic revenue implies that total opportunistic continuation revenue from $T - 1$ increases strictly with p . Both types' continuation revenues exceed the discounted sum of known-type revenues.

Suppose these continuation properties hold from $t + 1$. The right side of (36) is positive and increases strictly with p . At each fixed current tax, neither current investment nor the gain from reneging depends on p . The same argument as in Proposition 4 gives a unique current optimum. It is the full-commitment tax when feasible and otherwise the unique binding tax above it. A higher reputation strictly lowers a binding optimum toward the full-commitment tax. Current revenue exceeds the known-type revenue and weakly increases with reputation. Adding the strictly increasing discounted opportunistic continuation preserves strict monotonicity of total opportunistic revenue.

An unexpected announcement gives either type at most the discounted sum of known-type revenues. Current revenue and both types' continuation revenues at the selected policy exceed these amounts. The two announcement incentives therefore hold. This completes the induction to $t = 1$. $\square$

A.11 Firm-Specific Taxation

Partition firms into a finite number of observable groups with fixed, positive population shares. Production parameters, endowments, and origin-country shifting costs can differ across groups. Each firm divides its own capital between the two countries and shifts profits only within its multinational group. Neither transfers nor capital constraints connect different firms. All groups face the same exogenous foreign tax and the same government type, with common prior p .

Allow the government to set one rate for each group. It maximizes total revenue using population shares as weights. The commitment type implements the entire announced policy, while the opportunistic type can change any of its rates after investment. Upon an unexpected policy announcement, every group assigns probability zero to commitment.

Proposition A.2 (Reputation with firm-specific taxation). *Suppose each group's profits have the form $\rho(k) = zk^\alpha - r^*k$, and its parameters satisfy the assumptions and four restrictions of Proposition 1. Among supported pooling policies, every policy maximizing the commitment type's revenue sets each group's rate equal to a solution of its separate commitment problem. As reputation rises from any $p > 0$ to any $p' \in (p, 1]$, every group's chosen tax strictly falls and its investment strictly rises. Both government types' announcement incentives hold for the policy as a whole.*

Proof. For each group, the origin-country specification gives a discretionary tax independent of its installed capital. Allowing a separate rate for each group makes the opportunistic government's maximization separable. Each firm therefore faces exactly

TABLE 7: Taxes under commitment across production curvature and shifting costs

Production curvature α	Shifting cost γ	Tax under commitment (%)		
		$p = 0.1$	$p = 0.5$	$p = 0.9$
0.50	0.50	34.6	33.2	31.9
0.50	1.25	50.1	46.8	44.0
0.50	4.00	67.4	60.4	54.9
0.66	0.50	32.4	30.0	27.9
0.66	1.25	45.2	40.3	35.9
0.66	4.00	59.5	49.7	41.9
0.80	0.50	29.6	25.8	22.4
0.80	1.25	39.9	32.3	26.3
0.80	4.00	51.9	35.8	28.3

Notes: Taxes are in percent. Profits are $\rho(k) = k^\alpha - 0.04k$, with $\bar{k} = 10$ and $\tau^* = 0.15$. The two displayed parameters are held fixed within each reputation comparison.

the investment and reporting problem in Proposition 1, evaluated at its own parameters and the common prior.

The commitment type's objective without announcement constraints is the weighted sum of group revenues. Since each weight is positive, a maximizing policy must maximize revenue separately in every group. By Proposition 1, such a choice gives each type at least its revenue in that group's known-opportunistic allocation. Summing these inequalities with the population weights gives the two constraints for the entire policy. An unexpected announcement cannot yield either type more than the same weighted sum of known-type revenues. Thus the unconstrained group optima satisfy the announcement constraints, and adding those constraints leaves the solution unchanged.

Apply Proposition 1 separately to the groups. It orders all their maximizing taxes and investment choices strictly across priors, which establishes the result for the joint policy. At $p = 1$, each group solves its full-commitment problem. $\square$

B Numerical Sensitivity and Solution Methods

This appendix examines production curvature, shifting costs, and firm heterogeneity, and describes the solution methods used in Sections 4.4 and 5.4.

B.1 Production Curvature

I change production curvature while retaining the baseline reporting technology. The tax under commitment declines with reputation in all nine combinations of production curvature and shifting costs.

TABLE 8: Tax policies and investment under alternative costs

Specification	Tax under commitment (percent)			Host investment		
	$p = 0.1$	$p = 0.5$	$p = 0.9$	$p = 0.1$	$p = 0.5$	$p = 0.9$
$\gamma = 0.5$	32.41	30.04	27.85	3.549	3.651	3.909
$\gamma = 1.25$ (baseline)	45.19	40.33	35.88	2.442	2.632	3.153
$\gamma = 1.5$	47.67	42.18	37.19	2.227	2.436	3.019
$\phi = 3$	39.80	36.09	14.46	3.308	3.407	4.709
$\gamma^* = 0.5$	39.59	36.47	33.43	3.012	3.139	3.465
$\gamma^* = 2$	49.24	42.96	37.46	2.040	2.266	2.935
$\tau^* = 0.3, \phi = 1$	50.69	44.77	40.05	2.787	3.086	3.837
$\tau^* = 0.3, \phi = 5$	28.68	28.53	28.36	5.118	5.125	5.145

Notes: Parameters not varied equal the baseline in Figure 2. Costs are deductible in both countries. The last two rows use a foreign tax of 0.3. The quadratic benchmark has $\phi = 1$ and equal home and foreign cost parameters. Every displayed allocation satisfies the independent firm, tax, and announcement-incentive checks.

B.2 Alternative Cost Functions and Foreign Taxes

To vary cost curvature, replace the host and foreign shifting costs in (4) by

$$\frac{\gamma\theta^{\phi+1}}{2\rho(k)^\phi} \quad \text{and} \quad \frac{\gamma\theta^{\phi+1}}{2\rho(\bar{k}-k)^\phi}, \quad \phi \in \{1, 3, 5\}. \quad (37)$$

The odd values of ϕ make both costs nonnegative for either direction of shifting. The baseline is quadratic in the transfer. The other choices give quartic and sixth-power costs. For a shifting share below one, increasing ϕ lowers cost as well as changing curvature. These comparisons therefore do not hold a reporting elasticity fixed. Both taxable bases must remain nonnegative under each cost function.

For asymmetric quadratic costs, retain the host parameter γ and replace the foreign parameter by γ^* . The unconstrained transfer becomes

$$\theta(k, \tau) = \frac{(\tau^* - \tau)\rho(k)\rho(\bar{k} - k)}{\gamma(1 - \tau)\rho(\bar{k} - k) + \gamma^*(1 - \tau^*)\rho(k)}. \quad (38)$$

If a base constraint binds, the firm chooses the nearest feasible transfer. Changing either cost affects reporting, investment, the tax after investment, and the announcement. The foreign-cost comparison fixes $\gamma = 1.25$ and uses $\gamma^* = 0.5, 1.25, 2$.

In Table 8, quartic costs allow the tax under commitment at high reputation to attract inward shifting. At $p = 0.9$, the transfer is 0.291 into the host, compared with 0.289 out of the host in the quadratic benchmark. A lower tax under commitment can therefore attract inward profit shifting as well as reduce outward shifting.

The sixth-power comparison at $\tau^* = 0.3$ also retains the declining tax under commitment. Its small response to reputation illustrates the quantitative sensitivity to the cost function.

TABLE 9: Host allocations as the foreign tax changes

Foreign tax	τ^R	τ^O	Host investment	Θ/Λ
2	36.73	42.37	2.351	-25.36
10	38.93	45.29	2.514	-21.82
30	44.77	53.20	3.086	-12.17
50	51.73	62.10	4.040	-1.59
80	69.34	77.46	6.747	13.65
98	94.90	94.90	8.654	21.27

Notes: All tax rates and the transfer share are in percent. Reputation is $p = 0.5$ and $\gamma = 1.25$. The transfer share is measured under commitment. Positive values denote inward shifting. The full comparison uses foreign taxes from 0.02 through 0.98 in increments of 0.02.

A higher foreign rate lowers the return on foreign investment and changes the host's optimal tax. Host investment rises with these two adjustments, and sufficiently high foreign taxes induce inward profit transfers. The exercise treats the foreign rate as given throughout.

B.3 Cost Treatment and Reporting Responses

The cost of profit shifting in this model is incurred when firms adjust transfers within an existing multinational structure. It does not model the cost of establishing that structure. Structural estimates distinguish these two kinds of cost (Bilicka et al., 2024; Altshuler et al., 2025), and McClure (2023) discusses operational costs alongside audit risk and financial-reporting incentives. I use the comparisons below to examine alternative tax treatment of the resource costs while matching the reporting response at a common allocation.

I measure the reporting response by the semi-elasticity of the host tax base, defined as minus the derivative of its log with respect to the tax rate at fixed capital. For bilateral deductible costs, evaluated at equal taxes and equal profits, the response is $1/[2\gamma(1 - \tau^*)]$. It equals 0.471 at the baseline parameters $\gamma = 1.25$ and $\tau^* = 0.15$. To consider larger responses, I use 1.26 from Kallen (2023) and 1.50 from Beer et al. (2018). Bilicka et al. (2024) also report a short-run aggregate estimate of 1.50 holding capital and tax-planning capacity fixed. I use these numbers as reference magnitudes for the exercise. They concern different populations and allocations, so matching them does not identify a structural shifting-cost parameter.

With non-deductible costs in both countries, the firm's reporting objective is

$$(1 - \tau)[\rho(k) + \theta] + (1 - \tau^*)[\rho(\bar{k} - k) - \theta] - \frac{\gamma\theta^2}{2} \left[\frac{1}{\rho(k)} + \frac{1}{\rho(\bar{k} - k)} \right]. \quad (39)$$

TABLE 10: Optimal taxes with alternative shifting costs

Costs incurred in	Tax deductible?	Tax-base semi-elasticity	Tax under commitment (%)		
			$p = 0.1$	$p = 0.5$	$p = 0.9$
Both countries	Yes	1.26	31.5	29.3	27.2
Both countries	No	1.26	38.1	33.8	30.2
Country from which profits are shifted	Yes	1.26	30.6	29.3	27.9
Both countries	Yes	1.50	29.2	27.3	25.6
Both countries	No	1.50	34.3	30.9	28.0
Country from which profits are shifted	Yes	1.50	28.5	27.3	26.2

Notes: Taxes are in percent. Each reference response holds capital fixed at equal tax rates and equal profits before shifting. Production parameters are the same within each comparison. Nonnegative tax bases apply throughout. The rows match responses at this reference allocation, not at every equilibrium.

The interior transfer is $(\tau^* - \tau)\rho(k)\rho(\bar{k} - k)/[\gamma(\rho(k) + \rho(\bar{k} - k))]$, subject to nonnegative bases. Its reference base semi-elasticity is $1/(2\gamma)$. With deductible costs only at the origin, the outward transfer instead follows (22), giving reference semi-elasticity $1/[\gamma(1 - \tau^*)]$. Inward shifting deducts costs abroad. In every case, the tax maximization compares all candidate optima, including zero, one, and points where a reporting constraint starts to bind.

Table 10 compares costs incurred in both countries with costs incurred only at the origin, with and without tax deductibility. Each of the six comparisons gives a lower optimal tax under commitment at a higher reputation. Matching the reporting response at a common allocation thus preserves the direction of the baseline result across these specifications.

B.4 Firms with Different Shifting Costs

To allow for differences in access to profit shifting, I divide multinationals into two groups with different origin-country shifting costs. Their production functions, capital endowments, and foreign tax rates are the same. A single host statutory rate applies to both groups, so the government must choose its tax using their combined reported profits. In particular, it cannot implement each group's preferred discretionary rate separately.

For each pair of cost parameters and population share, I solve both groups' investment choices together with the government's common discretionary tax. I then find the commitment type's preferred announcement subject to the announcement incentives of both types. The revenue following an unexpected announcement is computed from the known-opportunistic equilibrium with this same composition of firms. This procedure allows each group's investment, as well as the tax on their combined base, to adjust to heterogeneity.

TABLE 11: The common tax under commitment with heterogeneous firms

Low cost	High cost	Low-cost share	$p = 0.1$	$p = 0.5$	$p = 0.9$
0.50	2.00	0.25	32.91	31.26	29.60
0.50	2.00	0.50	28.59	27.42	26.22
0.50	2.00	0.75	25.61	24.70	23.76
0.75	1.75	0.25	34.86	33.02	31.12
0.75	1.75	0.50	32.01	30.51	28.97
0.75	1.75	0.75	29.76	28.51	27.21
1.00	1.50	0.25	35.15	33.29	31.36
1.00	1.50	0.50	33.78	32.09	30.34
1.00	1.50	0.75	32.56	31.02	29.42

Notes: The last three columns report tax rates in percent. The first two columns give the groups' shifting-cost parameters. The third is the fraction of firms with the lower cost. All firms have $z = 1$, $\alpha = 0.66$, $r^* = 0.04$, and $\bar{k} = 10$, and face $\tau^* = 0.15$. Costs are deductible in the country of origin. Both groups' investments and the common discretionary tax are jointly determined.

Every cost and population-share comparison in Table 11 gives a lower tax under commitment at a higher reputation. The same ordering holds at the intermediate priors $p = 0.3$ and 0.7 . Thus the baseline mechanism operates in these calculations even though firms differ in the protection obtained from shifting and the government must tax them at a common rate.

For each reported allocation, I check both groups' investment conditions and compare the government's revenue at the endpoints, reporting-constraint transitions, and interior maxima. As a separate check, setting the two costs equal reproduces the homogeneous origin-country model. These calculations verify the reported equilibria without imposing uniqueness of an equilibrium in the heterogeneous model.

B.5 Longer Horizons and Changes in Shifting Costs

I implement the finite-horizon extension in Appendix A.10. The government and reporting technology are unchanged across horizons. Backward induction chooses each preterminal tax subject to the revenue loss after a deviation. Both types maintain every preterminal policy, while terminal investment allows for discretionary revision.

In Table 12, the government can sustain a lower initial rate as its reputation improves. With enough periods in this illustration, it reaches the full-commitment rate at all three priors. Extending the horizon gives the government additional cohorts from which it can collect revenue, increasing what it would sacrifice by revealing its type. The ordering across reputation levels follows Proposition A.1, while the number of periods needed to attain full commitment depends on the parameters used in the table.

For Table 6, I recompute the terminal-period equilibrium at each shifting cost, first with the prior p and then with the opportunistic type known. These two calculations

TABLE 12: The initial tax over different finite horizons

Periods	$p = 0.1$	$p = 0.5$	$p = 0.9$
2	34.29	32.43	30.49
3	33.29	31.37	30.32
5	31.11	30.32	30.32
10	30.32	30.32	30.32
30	30.32	30.32	30.32
Full commitment	30.32	30.32	30.32

Notes: Taxes are in percent. Costs are deductible in the country of origin. Parameters are $z = 1$, $\alpha = 0.66$, $r^* = 0.04$, $\bar{k} = 10$, $\tau^* = 0.15$, $\gamma = 1.25$, and $\beta = 0.6$. The horizon counts the initial period through the terminal period. Each column holds initial reputation fixed.

determine the continuation loss in (13). I next solve first-period investment under certain implementation and find the most profitable revision at that installed capital. The reported revenue changes evaluate the two sides of the constraint at the original tax, with investment and reporting adjusted to the new cost. The final tax comparison reoptimizes the initial policy as well.

I also vary government patience and the initial level of shifting costs. The eighteen comparisons combine the two cost changes, three discount factors, and three reputation levels. Current and future revenue changes are evaluated at the original optimal sustainable tax. The sustainable and full-commitment tax increases each compare the reoptimized policies. Their difference measures the additional tax increase associated with the implementation constraint. It is zero when both policies attain the full-commitment allocation.

TABLE 13: Shifting costs, patience, and the incentive to maintain policy

β	p	Change in revenue		Tax increase (pp)		
		Current gain	Future loss	Sustainable	Full commitment	Difference
$\gamma = 0.5$ to $\gamma = 1.25$						
0.3	0.1	0.09375	0.00005	10.98	8.65	2.33
0.3	0.5	0.10250	0.00149	10.25	8.65	1.59
0.3	0.9	0.11118	0.00502	9.47	8.65	0.82
0.6	0.1	0.09464	0.00011	10.91	8.65	2.26
0.6	0.5	0.10688	0.00298	9.86	8.65	1.21
0.6	0.9	0.11885	0.01005	8.75	8.65	0.10
0.9	0.1	0.09532	0.00016	10.86	8.65	2.21
0.9	0.5	0.11019	0.00447	9.57	8.65	0.92
0.9	0.9	0.11969	0.01507	8.65	8.65	0.00
$\gamma = 1.25$ to $\gamma = 1.5$						
0.3	0.1	0.00564	0.00002	2.41	1.66	0.75
0.3	0.5	0.01017	0.00065	2.17	1.66	0.51
0.3	0.9	0.01459	0.00218	1.92	1.66	0.26
0.6	0.1	0.00609	0.00005	2.39	1.66	0.73
0.6	0.5	0.01240	0.00130	2.05	1.66	0.39
0.6	0.9	0.01841	0.00437	1.69	1.66	0.03
0.9	0.1	0.00644	0.00007	2.37	1.66	0.71
0.9	0.5	0.01407	0.00195	1.95	1.66	0.29
0.9	0.9	0.01888	0.00655	1.66	1.66	0.00

Notes: Costs are deductible in the country of origin. Parameters are $z = 1$, $\alpha = 0.66$, $r^* = 0.04$, $\bar{k} = 10$, and $\tau^* = 0.15$. Current gain and future loss are changes in the two sides of (13), evaluated at the original optimal tax, with investment and continuation equilibria recomputed at the new cost. Future loss includes β . Revenue is in model units. The last three columns are percentage-point changes in optimal tax rates. Difference is the sustainable tax increase minus its full-commitment counterpart. Every allocation satisfies the announcement and implementation constraints.

B.6 Numerical Solution and Verification

I solve the investment and discretionary-tax problems jointly at each announcement, then maximize the commitment type's revenue over announcements that satisfy both types' incentives. Individual firms take the implemented tax as given. To find the best discretionary tax, I compare revenue at zero and one, at every transition into a binding reporting constraint, and at interior maxima between transitions.

In the two-period problem, these one-period solutions supply the continuation revenues under belief p and after revelation of the opportunistic type. For each candidate initial tax, I solve investment assuming certain implementation and evaluate the most profitable revision at that capital. The chosen tax maximizes current revenue subject to (13) and (14). Figure 4 repeats the calculation for $\beta = 0.3, 0.6, 0.9$.

I check the reported allocations with separate maximizations over investment, reporting, and implemented taxes, and use a finer announcement search to check

the policy choice. The replication files report the residuals and these independent comparisons. They also verify that the quadratic cost specification reproduces the baseline payoff function.

At $z = 1$, $\alpha = 0.66$, $r^* = 0.04$, $\bar{k} = 10$, and $\tau^* = 0.15$, all four restrictions in (9) hold for $\gamma = 0.5$ and 1.25. The baseline parameters also satisfy the no-shifting restriction in Appendix A.7. The replication files document separate symbolic and numerical checks.

For the alternative cost functions and foreign tax rates, I search for discretionary optima on both sides of the foreign rate. This permits inward as well as outward shifting. I then separately maximize over the tax and individual investment on continuous domains to check each allocation reported in the paper.

C Data and Additional Empirical Results

This appendix describes the samples used in the empirical analysis and reports the complete regression tables. The replication files include the ICRG component scores, the observations used for estimation, and records of the additional specifications considered.

C.1 Sources and Estimation Samples

I obtain the monthly component scores from ICRG Table 3B, January 1984–December 2021. Within a country, I require twelve reported months to construct an annual mean, and I merge the annual index with outcomes dated one year later. I treat historical aggregates separately from successor countries. Of the 5,006 annual observations matched to the public January 2026 QoG series, 5,005 have identical values. Indonesia in 2018 is the exception, with 0.472222 from the component scores and 0.694444 in the public series. The regression data use the former. Both are preserved in the source comparison.

For Tax Foundation I obtain 2,432 usable country-year observations in 136 countries. The OECD series gives 1,907 observations in 101 countries, of which 1,865 country-years in 101 countries can also be matched to Tax Foundation. I form country means independently in the available and matched samples. Within either sample, a country's tax and explanatory variables are averaged over the same years. A country's presence in the FDI data is not required.

The FDI variable is `di_inflows` in the February 2021 Forbes–Warnock release, expressed in billions of current US dollars. I add four reported quarterly amounts to form an annual observation and retain positive totals, since the regression uses logs. A

country-year must also have the lagged quality-of-government index and the control variables. This leaves 56 countries and 902 observations during 2003–2019. The combined Belgium–Luxembourg series is excluded because it has no single corresponding quality-of-government score. Real GDP per capita is measured in constant 2015 dollars. Income and population are logged, and resource rents are expressed as a percentage of GDP.

The shifting estimates use OECD country-by-country reports for 2016 supplemented with US reports for 2017 (Garcia-Bernardo and Jansky, 2021). I reverse the sign of their outward-loss estimates to obtain net inward shifting and match the estimates to 2016 covariates. There are 116 countries, 85 of them with net outward shifting. The regressions treat the published outcome estimates as data when computing HC3 standard errors.

C.2 Inference and Supporting Comparisons

For the tax and profit-shifting coefficients, inference uses HC3 standard errors and the residual degrees of freedom. I cluster the FDI standard errors by country and use 55 degrees of freedom. I selected the index and examined specifications in an exploratory analysis. The reported p -values are conventional ones and contain no adjustment for that selection. The replication records include the other specifications examined.

The conditional tax comparison remains positive when havens are excluded. The coefficients are 8.411 ($p = 0.070$) for Tax Foundation and 14.560 ($p = 0.021$) for OECD. With continent effects, they are 11.160 ($p = 0.031$) and 20.084 ($p = 0.011$). These are checks on the relationship between country means after controlling for economic characteristics. Neither unconditional means nor annual within-country variation gives the same established relationship.

Clustering the controlled FDI regression by country and year gives $p = 0.058$. A restricted wild-cluster bootstrap with 999 draws gives $p = 0.071$. Omitting countries one at a time leaves every coefficient positive. The corresponding omissions in the profit-shifting regression leave the coefficient positive and significant at 5 percent.⁷

C.3 Full Versions of Tables

Tables 14–16 include the control coefficients omitted from the compact main tables. They reproduce Tables 1, 2, and 3, with identical observations and standard errors.

⁷Removing all havens leaves 49 FDI countries, with a coefficient of 1.976 (SE 1.227, $p = 0.114$). Among 99 non-haven countries, the profit-shifting coefficient is 3.274 ($p = 0.053$) with baseline controls and 4.116 ($p = 0.153$) after adding income and population. Thus the main estimates apply to the sample including havens.

TABLE 14: Quality of government and statutory corporate tax rates

	(1)	(2)	(3)	(4)
Quality of government	8.844* (4.615)	17.767*** (6.328)	13.326** (6.192)	16.459*** (6.145)
Log GDP per capita	−2.130*** (0.647)	−3.257*** (1.124)	−2.973*** (1.003)	−3.561*** (0.989)
Log population	1.367*** (0.457)	1.914*** (0.534)	1.642*** (0.524)	1.758*** (0.524)
Resource rents/GDP	0.021 (0.070)	0.002 (0.084)	−0.002 (0.082)	−0.020 (0.080)
Constant	17.012* (9.041)	13.194 (12.766)	17.879 (11.313)	19.718* (11.259)
Source	TF	OECD	TF	OECD
Country-year coverage	Available	Available	Matched	Matched
Income, population, rents	Yes	Yes	Yes	Yes
Comparison	Means	Means	Means	Means
Observations	136	101	101	101
Countries	136	101	101	101
R^2	0.201	0.286	0.263	0.311

Notes: Full version of Table 1. Each observation is a country mean over available 2003–2021 years. QoG is lagged before averaging and ranges from 0 to 1. Rates and rents/GDP are in percentage points. TF denotes Tax Foundation. Columns (3)–(4) use identical country-year observations. Parentheses contain HC3 standard errors. ***, **, and * denote significance at 1, 5, and 10 percent.

TABLE 15: Quality of government and annual inward FDI

	(1)	(2)
Quality of government	2.586* (1.334)	2.131* (1.157)
Log GDP per capita	–	1.563*** (0.341)
Log population	–	3.763*** (0.632)
Resource rents/GDP	–	0.029 (0.019)
Income, population, rents	No	Yes
Country effects	Yes	Yes
Year effects	Yes	Yes
Years	2003–2019	2003–2019
Observations	902	902
Countries	56	56
Within R^2	0.009	0.097

Notes: Full version of Table 2. The outcome is log positive annual inward FDI from Forbes–Warnock, summed over four quarters. QoG is lagged one year and ranges from 0 to 1. Both columns use identical country-year observations. Within R^2 removes country and year effects. Parentheses contain country-clustered standard errors. ***, **, and * denote significance at 1, 5, and 10 percent.

TABLE 16: Quality of government and estimated net inward profit shifting

	(1)	(2)
Quality of government	9.967*** (3.471)	9.995** (4.119)
Direct-investment controls	−1.526 (1.522)	−1.636 (1.603)
GDP growth	0.172 (0.172)	0.165 (0.220)
Manufacturing/GDP	0.019 (0.112)	0.002 (0.101)
Statutory tax rate	−0.064 (0.071)	−0.077 (0.120)
Log GDP per capita	−	0.034 (0.701)
Log population	−	0.188 (0.533)
Constant	−4.499 (3.324)	−7.283 (6.347)
Capital controls, growth, manufacturing	Yes	Yes
Statutory tax rate	Yes	Yes
Income and population	No	Yes
Year	2016	2016
Observations	116	116
Countries	116	116
R ²	0.177	0.179

Notes: Full version of Table 3. Both columns use the same 116 countries. Net inward shifting and manufacturing are percentages of GDP. Growth and tax rates are in percent. QoG and controls refer to 2016. QoG ranges from 0 to 1. Parentheses contain HC3 standard errors conditional on the published shifting estimates. ***, **, and * denote significance at 1, 5, and 10 percent.

C.4 Treaty Exposure and Investment

The investment series is V15110 in Eurostat’s inward FATS data. It records gross investment in tangible goods, or annual additions to tangible assets, in millions of euros. These are expenditures by foreign affiliates rather than an FDI flow or a stock of assets. I organize the observations by host and ultimate controlling parent’s country. The treatment group consists of pairs in the 2020 intra-EU termination agreement with a treaty in force immediately before the judgment in 2018. Such treaties in the sample were all in force by 2010. Both countries must fall under the agreement, and I remove pairs with a previously terminated treaty instead of using them as controls.

The estimating equation for host h , parent country j , and year t is

$$\begin{aligned} \mathbb{E}[\text{Investment}_{hjt} \mid h, j, t] = & \exp\left(\alpha_{hj} + \lambda_{ht} + \lambda_{jt} + \delta_{hj}(t - 2017) \right. \\ & \left. + \sum_s \beta_s \text{Covered treaty}_{hj} \mathbf{1}\{t = s\}\right). \end{aligned} \quad (40)$$

The PPML estimates of (40) use investment amounts, including reported zeros. There is no pair trend in the first column ($\delta_{hj} = 0$), and all annual effects are measured relative to 2017. A second normalization is needed once pair trends are included, since a linear combination of the annual indicators can reproduce a trend. I choose 2010 as the additional reference year. The coefficients in that column consequently describe departures from the line through the 2010 and 2017 reference points.

Identification comes from differences in treaty coverage across pairs after accounting for pair, host-year, and parent-year effects. The interpretation as an effect of the judgment requires that no concurrent change in investment, correlated with coverage, be left unexplained by those effects and the specified trends. Although hosts could choose their treaties in earlier years, the 2018 ruling itself was not their response to contemporaneous investment. Measurement of coverage raises a further issue. Eurostat assigns an affiliate to the country of its ultimate parent, but a treaty claim may use the nationality of an intermediate holding company instead.

The no-trend estimates use 2,622 observations from 349 pairs among 22 hosts and 22 parent countries. For the estimates with trends, these counts fall to 2,547 observations and 320 pairs. I obtain the smaller estimation samples by removing separated observations in PPML, such as groups reporting zero investment in every year. Pair trends lead to additional separation. The omitted observations are recorded in the replication files, and I check the score conditions for the observations used in estimation. A negative investment report is omitted before estimation and is never recoded to zero.

I cluster coefficient standard errors by host and parent country and use 21 degrees of freedom for inference. The two-way covariance matrix is not positive definite

TABLE 17: Annual investment coefficients around the Achmea judgment

	(1)	(2)
2010	−0.344 (0.271)	–
2011	−0.026 (0.284)	0.242 (0.314)
2012	−0.100 (0.181)	0.182 (0.219)
2013	−0.173 (0.257)	0.059 (0.235)
2014	−0.278* (0.138)	−0.102 (0.150)
2015	−0.172 (0.203)	−0.049 (0.127)
2016	−0.234*** (0.080)	−0.137 (0.111)
2018	−0.695*** (0.215)	−0.919*** (0.260)
2019	−0.253 (0.251)	−0.376 (0.346)
Pair-specific trends	No	Yes
Observations	2,622	2,547

Notes: Full annual treatment coefficients for the two regressions in Table 4. All columns include pair, host-year, and parent-year effects. Annual coefficients are relative to 2017. Column (2) also omits 2010 because pair trends absorb a linear path of treatment coefficients. Dashes denote terms not estimated. Parentheses contain standard errors clustered by host and parent country. ***, **, and * denote significance at 1, 5, and 10 percent.

for the joint tests of pre-event coefficients, so I calculate those tests using the valid host-clustered covariance and state this in the table. Without trends, the null of zero pre-event coefficients is rejected ($p = 0.017$). With trends, the test on the six remaining pre-event coefficients gives $p = 0.549$. This latter test concerns differences left over after fitting the pair trends.

Missing observations are concentrated in 2018. There are 352 reported pairs in 2017, 99 in 2018, and 334 in 2019. Requiring a complete ten-year panel would leave only six hosts, so I use the available observations as the main sample. The replication archive includes the balanced-panel and alternative-estimator comparisons. Throughout, the event is a change in treaty arbitration protection. It is not coded as a change in QoG or as an observed corporate-tax reversal.